

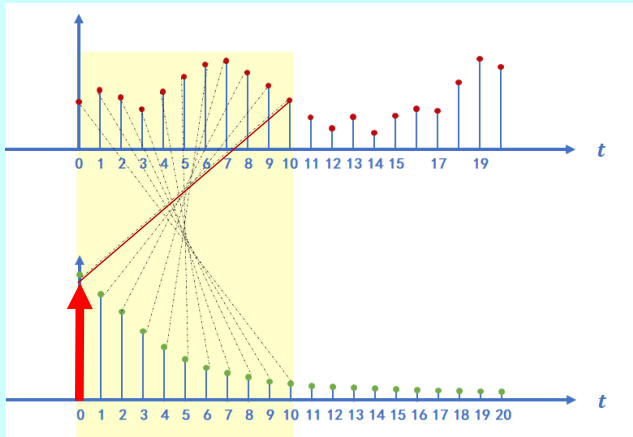
信号处理原理-03

刘华平
清华大学

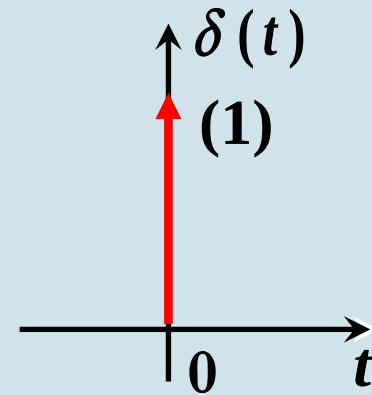
上节回顾

1

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$



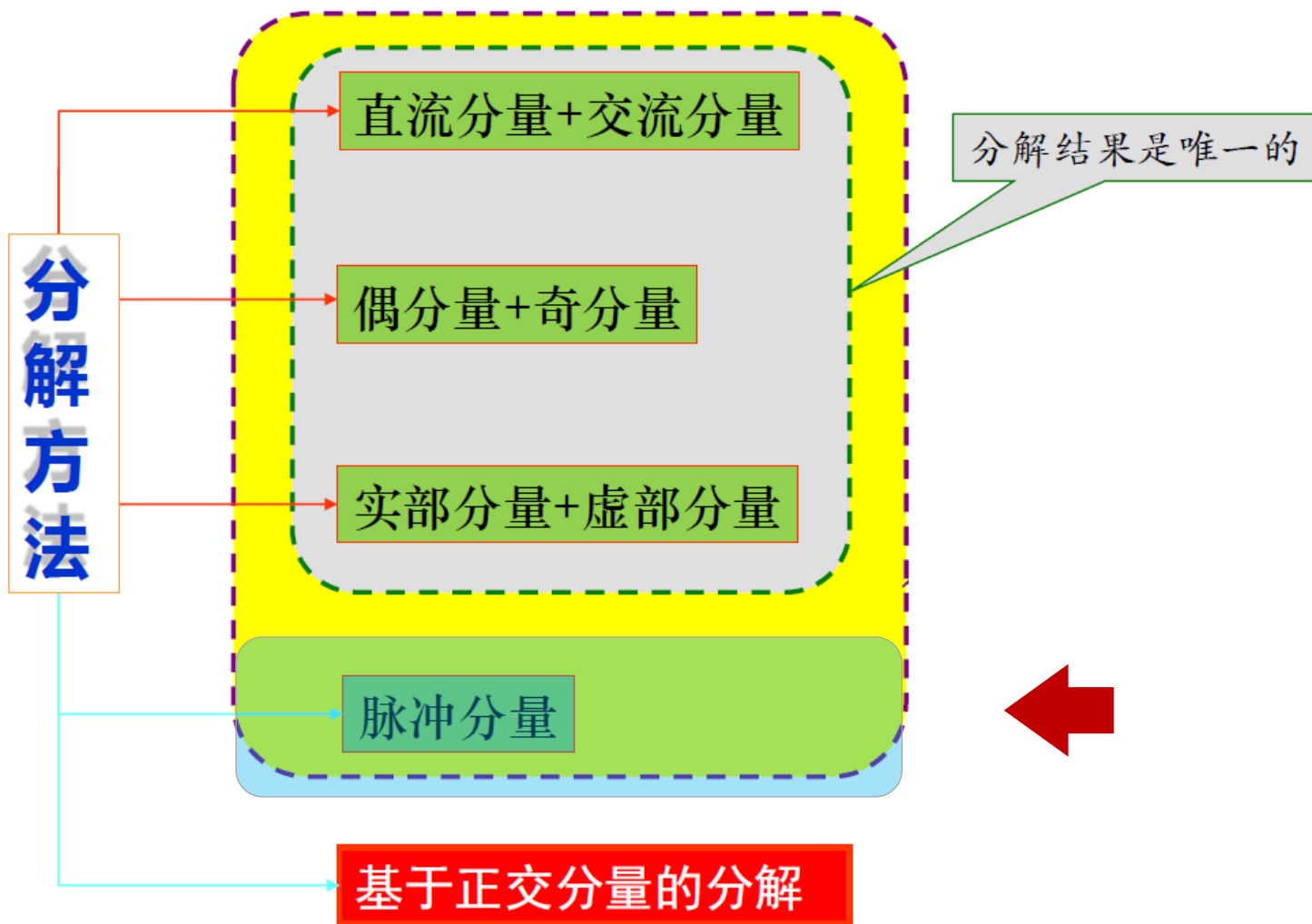
$$\begin{cases} \delta(t) = 0 & t \neq 0 \\ \delta(t) = \infty & t = 0 \\ \int_{-\infty}^{\infty} \delta(t)dt = 1 \end{cases}$$



$$f(t) * \delta(t) = f(t)$$

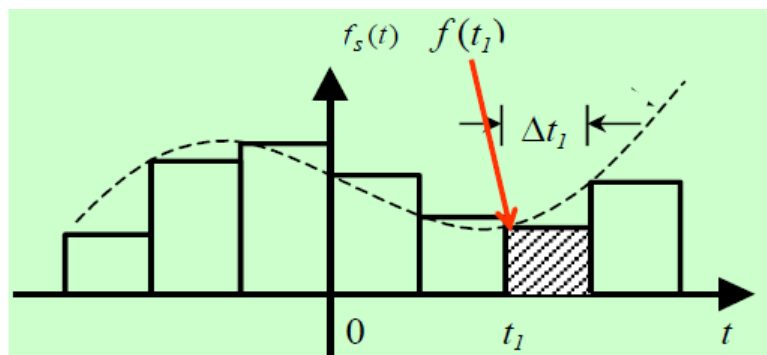
信号的基本分解

- 信号分解方法



➤ 脉冲分解

- 信号的脉冲分解：信号可以近似表示为一组矩形脉冲的和的形式

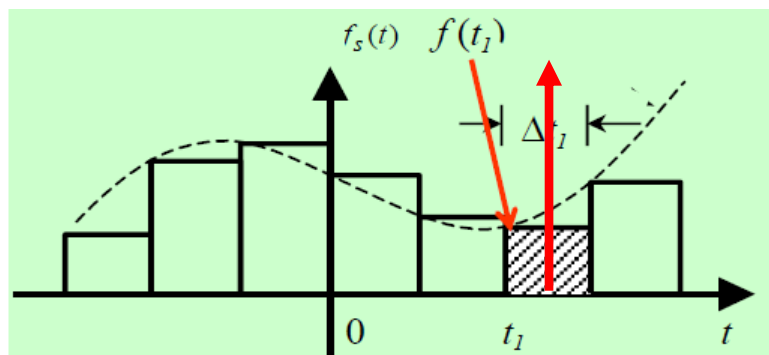


- 其中 t_1 处的矩形脉冲可以表示为：

$$f_{t_1}(t) = f(t_1)[u(t - t_1) - u(t - t_1 - \Delta t_1)]$$

➤ 脉冲分解

- 信号的脉冲分解：信号可以近似表示为一组矩形脉冲的和的形式

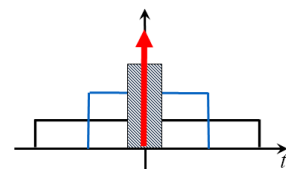


- 其中 t_1 处的矩形脉冲可以表示为：

$$f_{t_1}(t) = f(t_1)[u(t - t_1) - u(t - t_1 - \Delta t_1)]$$

- 原始函数可以表示为：

$$f(t) \approx \sum_{t_1=-\infty}^{\infty} f_{t_1}(t) = \sum_{t_1=-\infty}^{\infty} f(t_1)[u(t - t_1) - u(t - t_1 - \Delta t_1)]$$



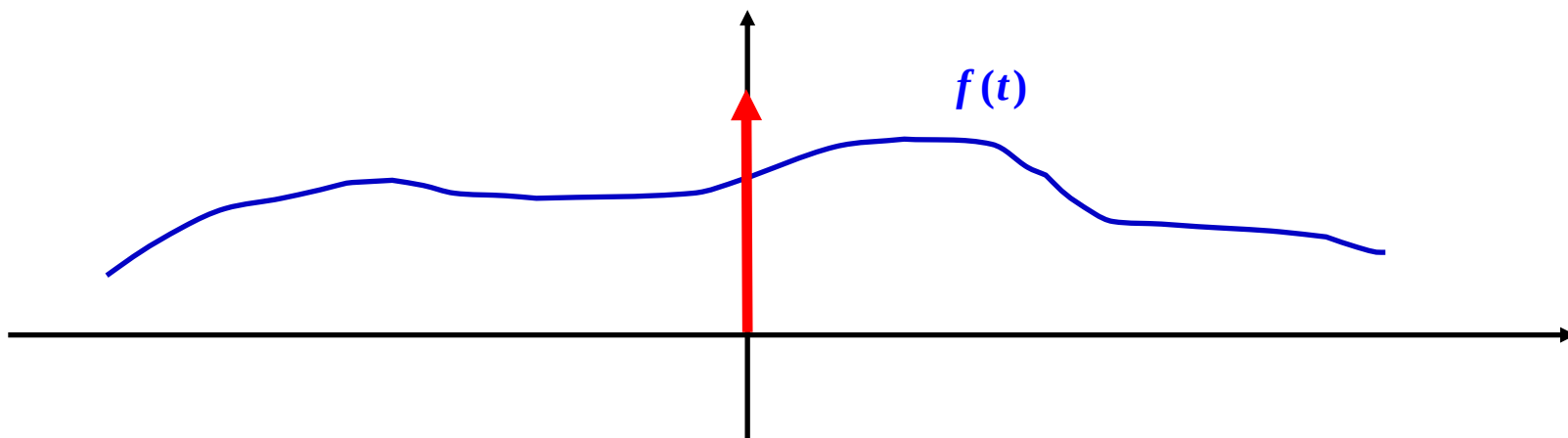
$$f(t) * \delta(t) = f(t) = \int_{-\infty}^{\infty} f(\tau) \cdot \delta(t - \tau) d\tau$$

➤ 脉冲分解

$$f(t) * \delta(t) = f(t) = \int_{-\infty}^{\infty} f(\tau) \cdot \delta(t - \tau) d\tau$$

||

$$f(t) * \delta(t)$$

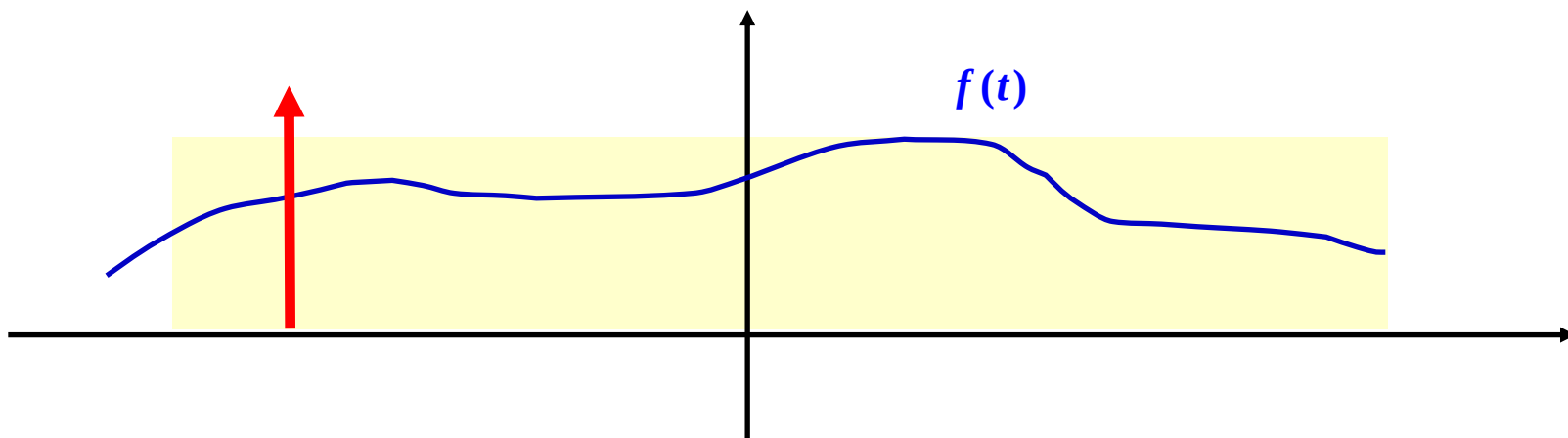


➤ 脉冲分解

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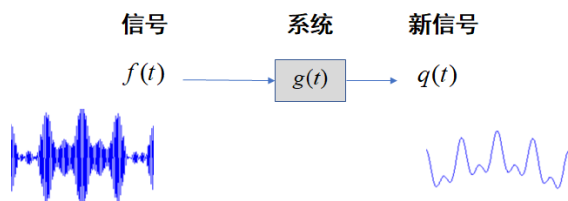
$$f(t) * \delta(t)$$



信号的基本分解——脉冲分解

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Why

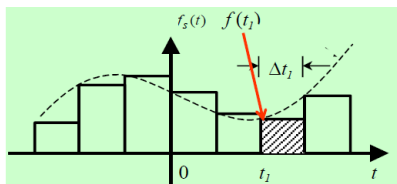


What

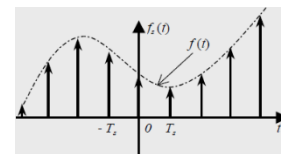
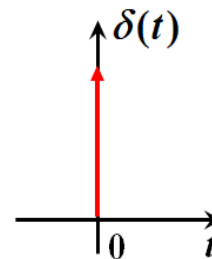
卷积计算

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

How

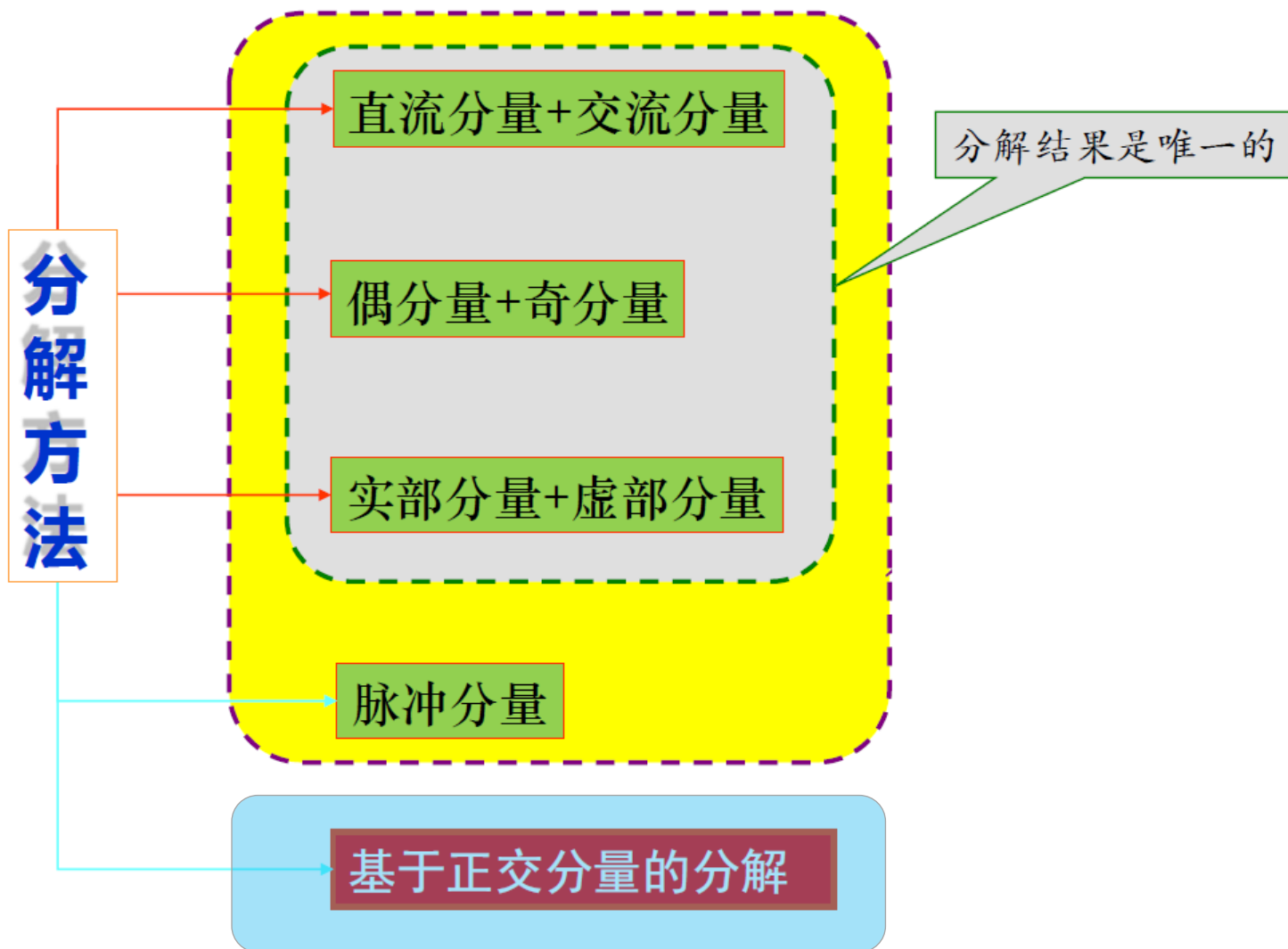


奇异信号



$$f(t) * \delta(t) = f(t) = \int_{-\infty}^{\infty} f(\tau) \cdot \delta(t - \tau) d\tau$$

➤ 信号分解方法



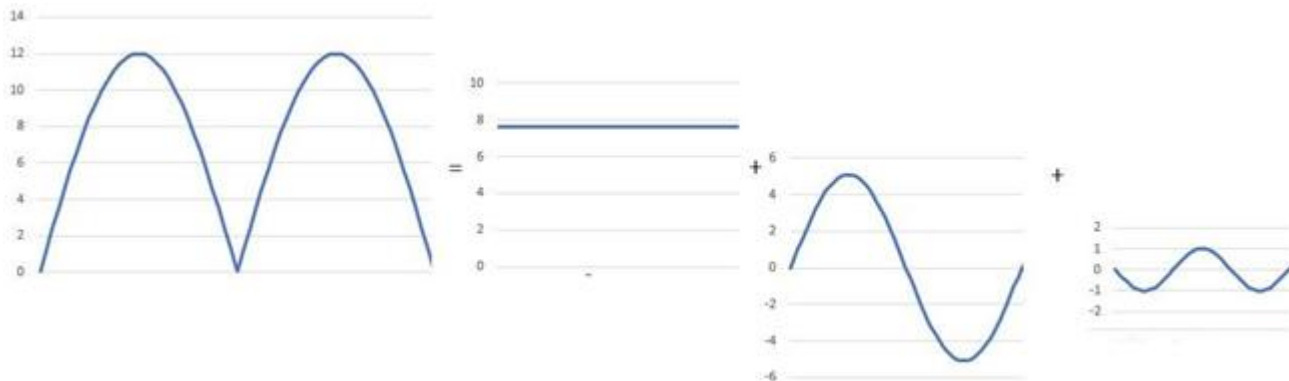
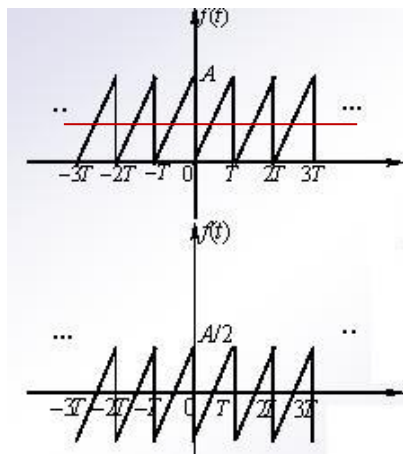
➤ 直流-交流分解

信号的直流分量

$$f_{DC}(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) dt \rightarrow \text{信号的均值}$$

信号的交流分量

$$f_{AC}(t) = f(t) - f_{DC}(t)$$



➤ 奇偶分解

信号的偶分量

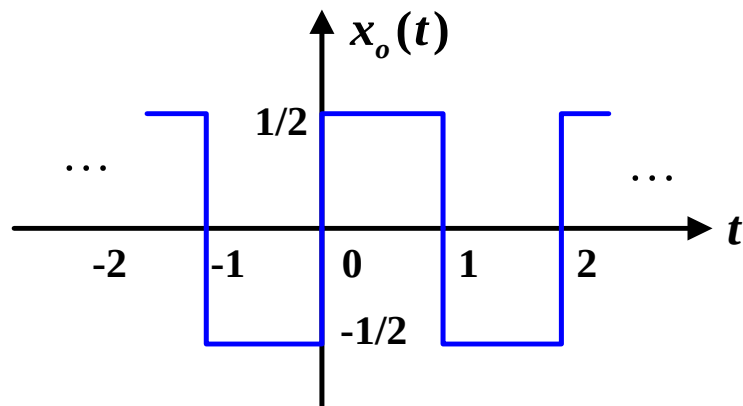
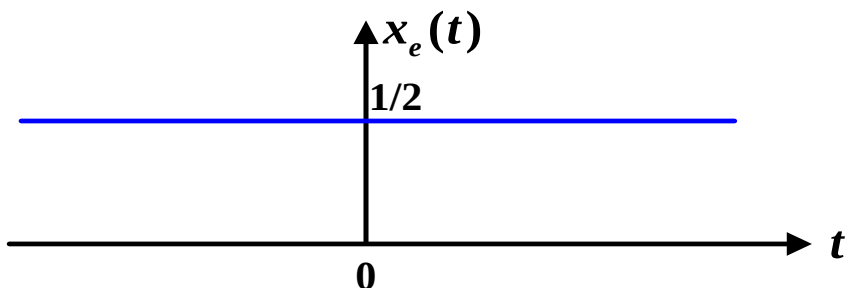
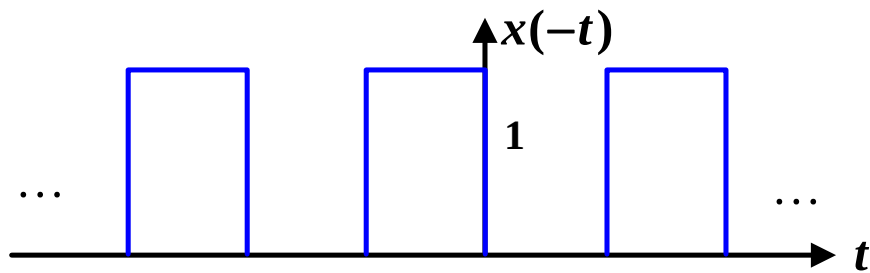
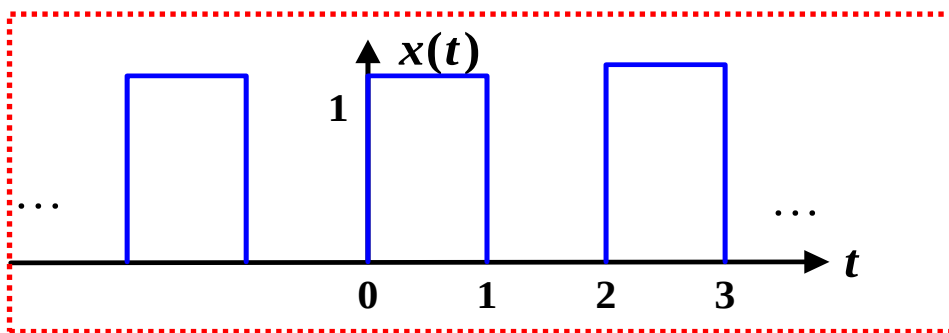
$$f_e(t) = Ev[f(t)] = \frac{f(t) + f(-t)}{2}$$

$$f_e(t) = f_e(-t)$$

信号的奇分量

$$f_o(t) = Od[f(t)] = \frac{f(t) - f(-t)}{2}$$

$$f_o(t) = -f_o(-t)$$



➤ 虚实分解

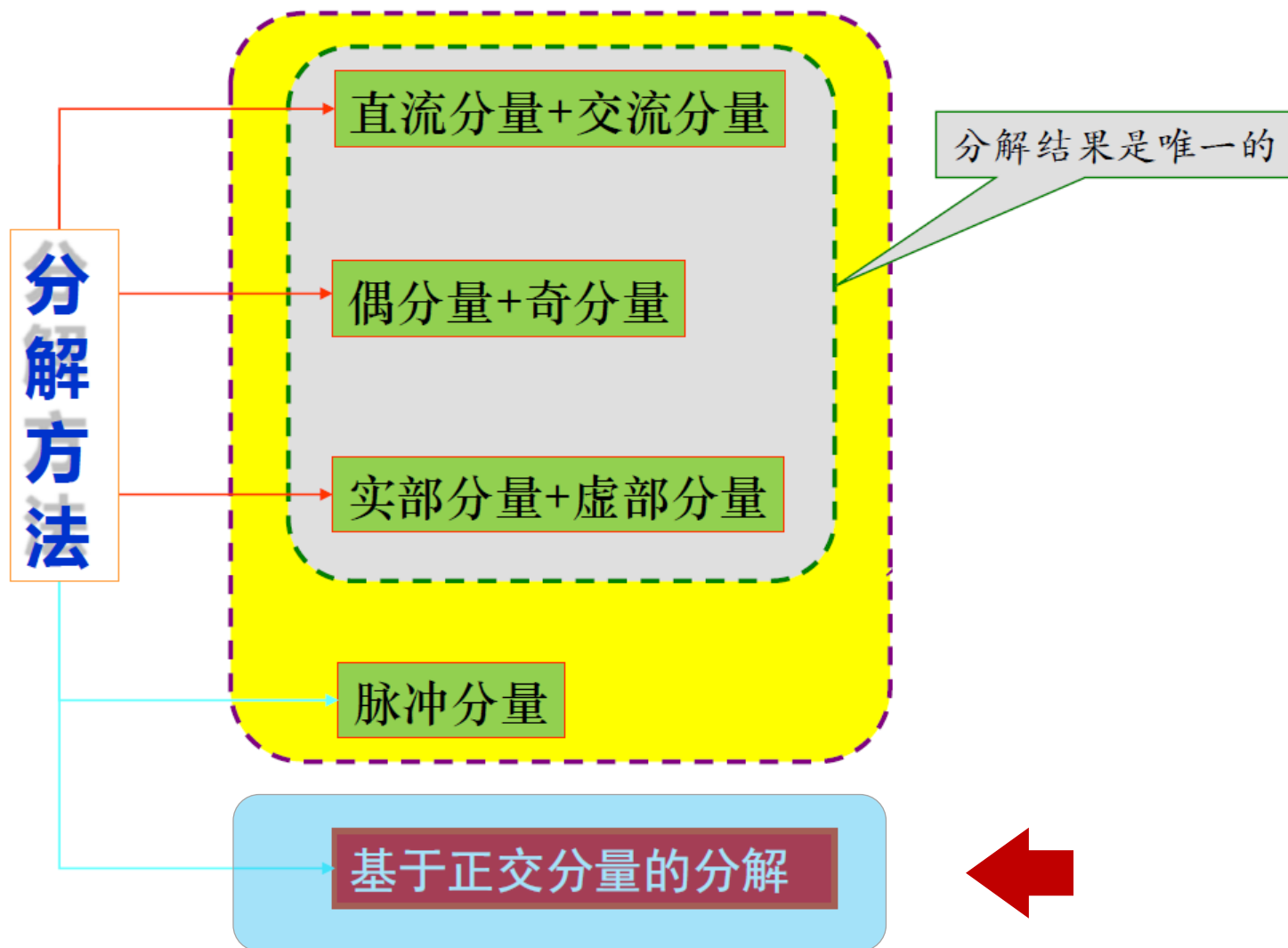
信号的实部分量

$$f_r(t) = \operatorname{Re}[f(t)] = \frac{1}{2}(f(t) + f^*(t))$$

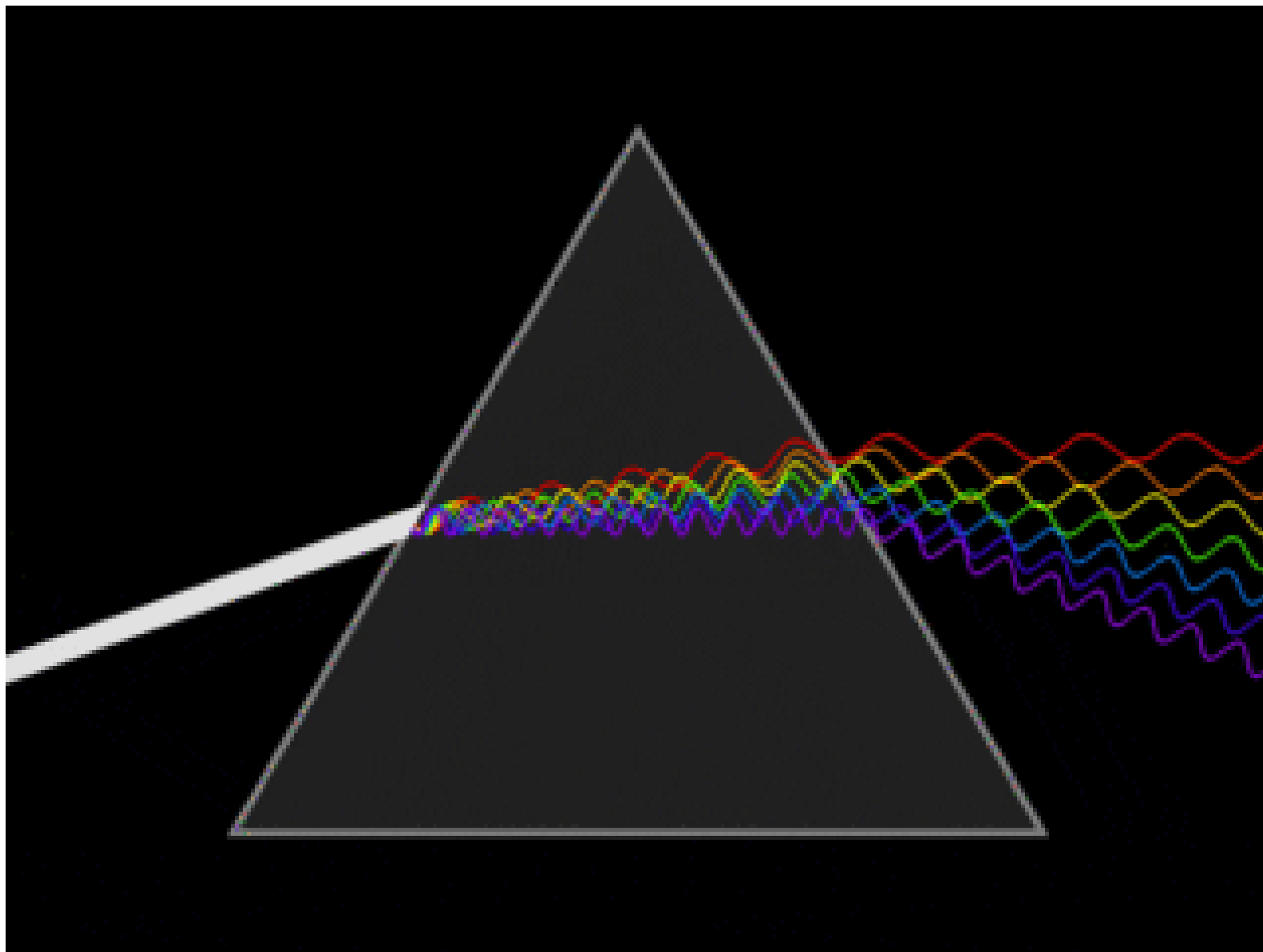
信号的虚部分量

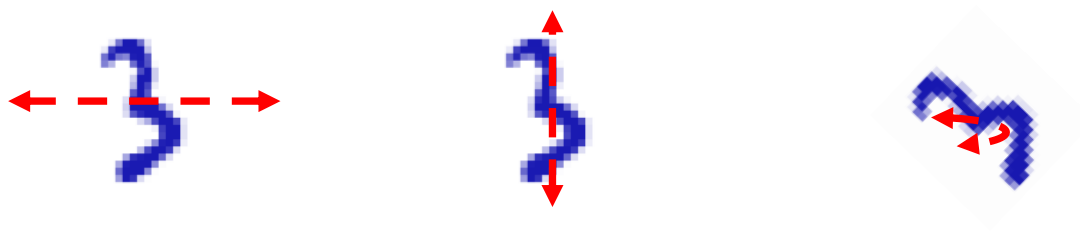
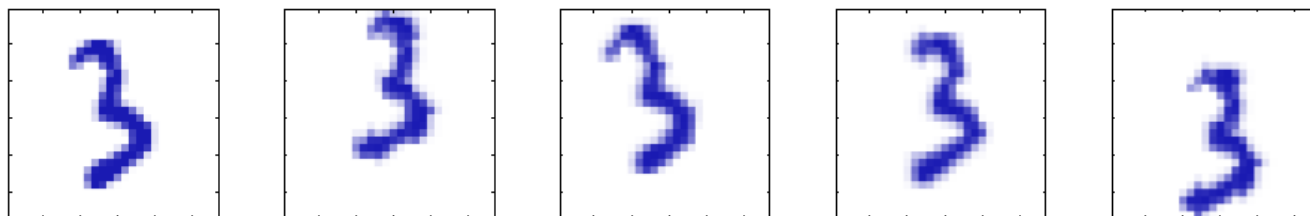
$$f_i(t) = \operatorname{Im}[f(t)] = \frac{1}{2j}(f(t) - f^*(t))$$

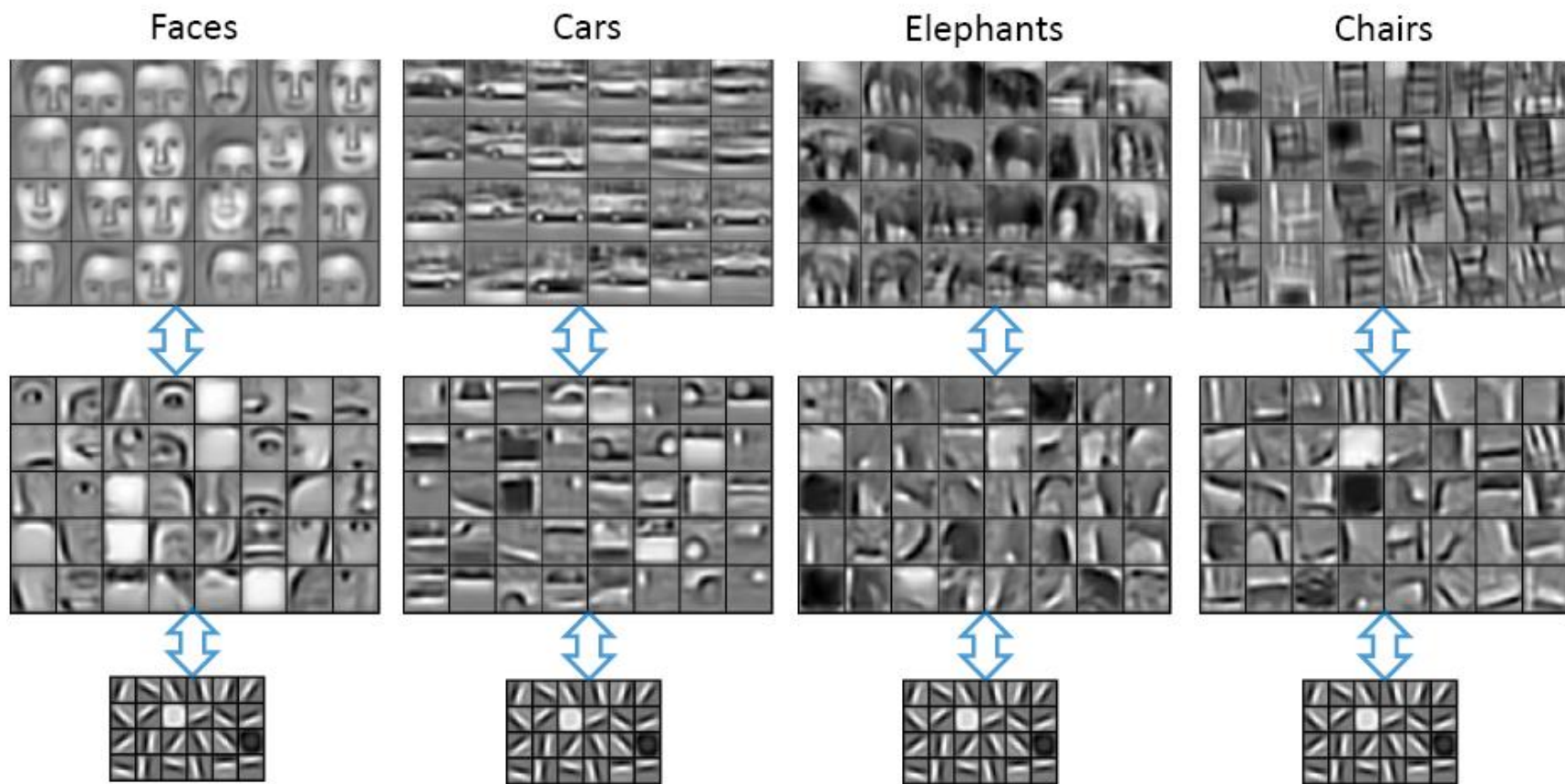
➤ 信号分解方法



信号的分解

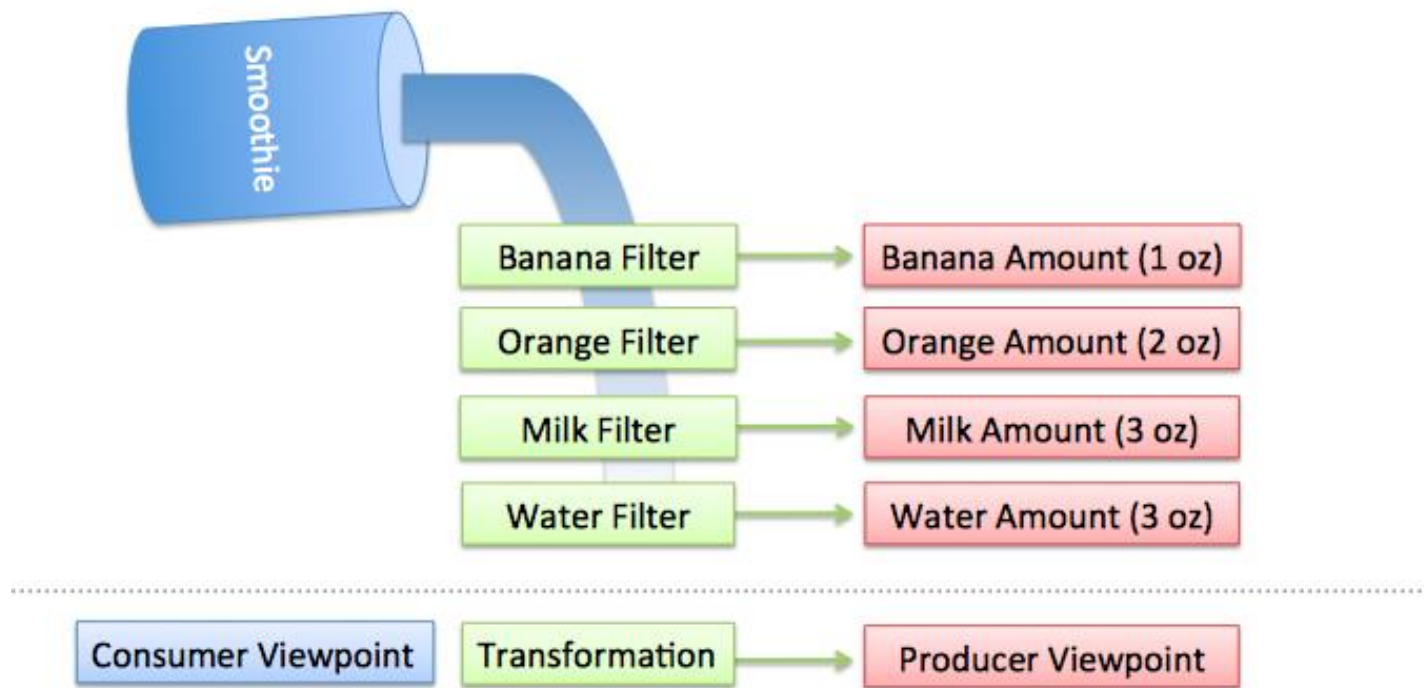






信号的分解

Smoothie to Recipe



完备

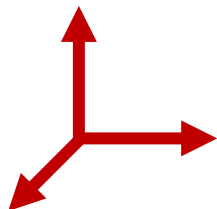
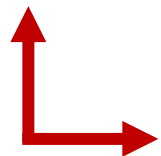
正交

信号的正交分解

向量的正交分解

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正交



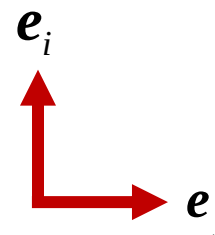
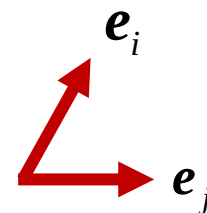
$\mathbf{e}_i$

$\mathbf{e}_j$

相互之间无法表示

$$\mathbf{e}_i^T \mathbf{e}_j = |\mathbf{e}_i| |\mathbf{e}_j| \cos \theta$$

$$\mathbf{e}_i^T \mathbf{e}_j = 0$$



向量的正交分解

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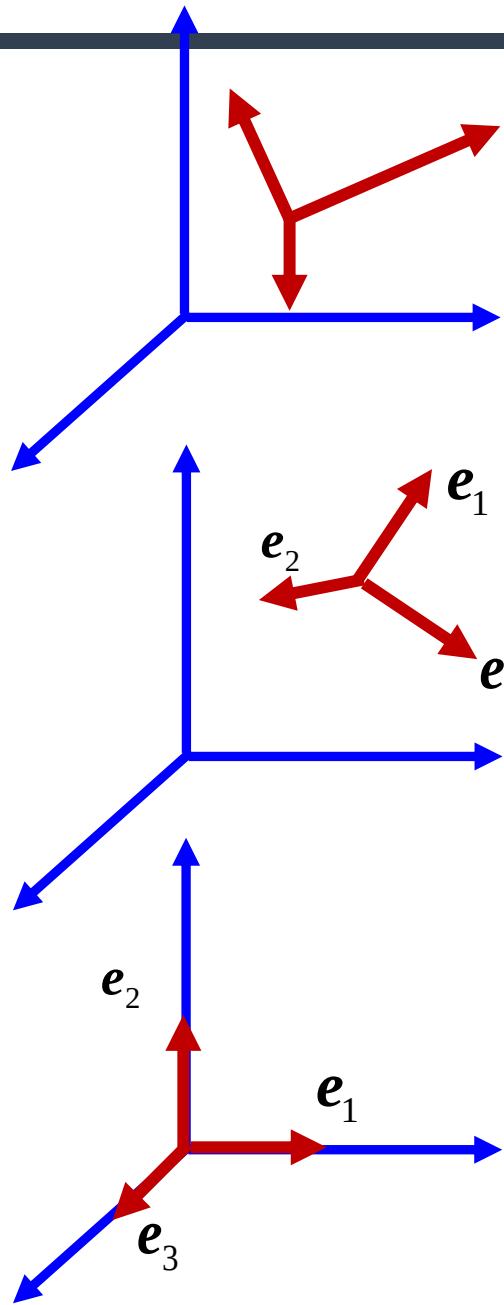
三维欧氏空间

➤ 向量基 $\mathbf{e}_i^T \mathbf{e}_j = 0 \quad i \neq j$

➤ 单位正交向量基

$$\mathbf{e}_i^T \mathbf{e}_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad \begin{array}{l} \text{(模为1)} \\ \text{(相互垂直)} \end{array}$$

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



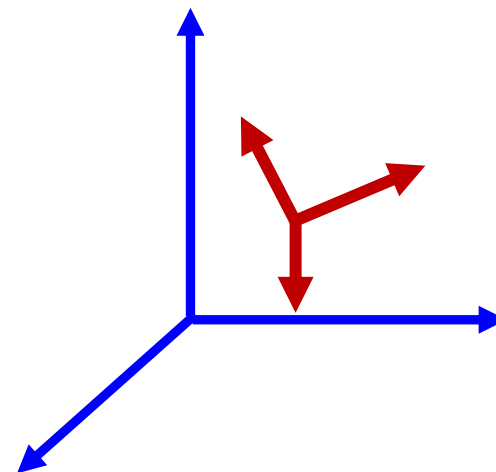
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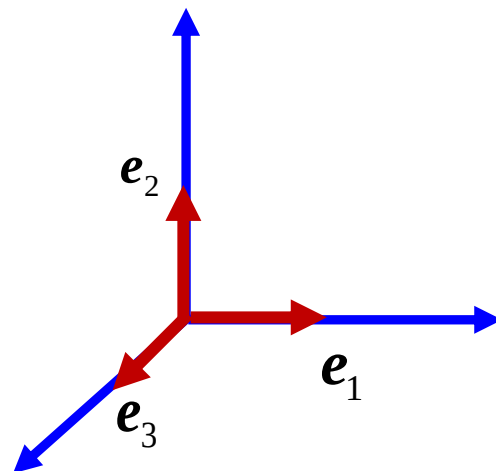


$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \quad \mathbf{a}_2 = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} \quad \mathbf{a}_3 = \begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix}$$

三维欧氏空间

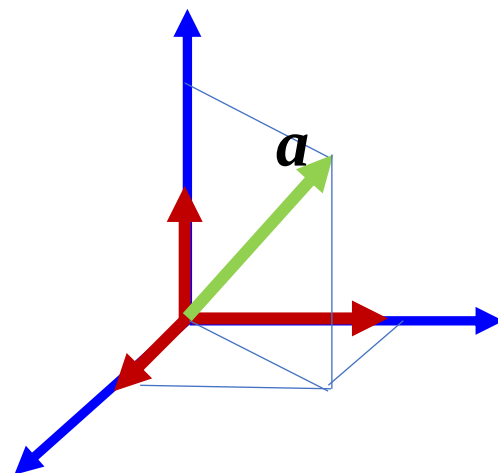
➤ 单位正交向量基

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



➤ 向量的正交分解

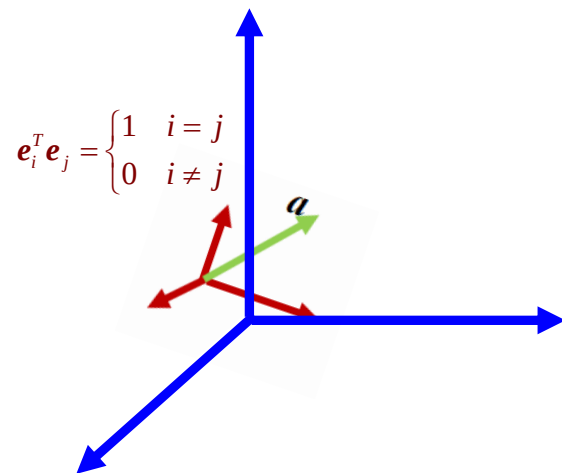
$$\mathbf{a} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 2\mathbf{e}_1 + 3\mathbf{e}_2 + \mathbf{e}_3$$



三维欧氏空间

➤ 向量的正交分解

$$\mathbf{a} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3 \quad \mathbf{a}^T = a_1 \mathbf{e}_1^T + a_2 \mathbf{e}_2^T + a_3 \mathbf{e}_3^T$$



$$\mathbf{a}^T \mathbf{e}_1 = a_1 \mathbf{e}_1^T \mathbf{e}_1 + a_2 \mathbf{e}_2^T \mathbf{e}_1 + a_3 \mathbf{e}_3^T \mathbf{e}_1$$

1 0 0

$$a_1 = \mathbf{a}^T \mathbf{e}_1 \quad a_2 = \mathbf{a}^T \mathbf{e}_2 \quad a_3 = \mathbf{a}^T \mathbf{e}_3$$

a_i 就是 $\mathbf{a}$ 在单位基向量 $\mathbf{e}_i$ 上的投影长度

向量的正交分解

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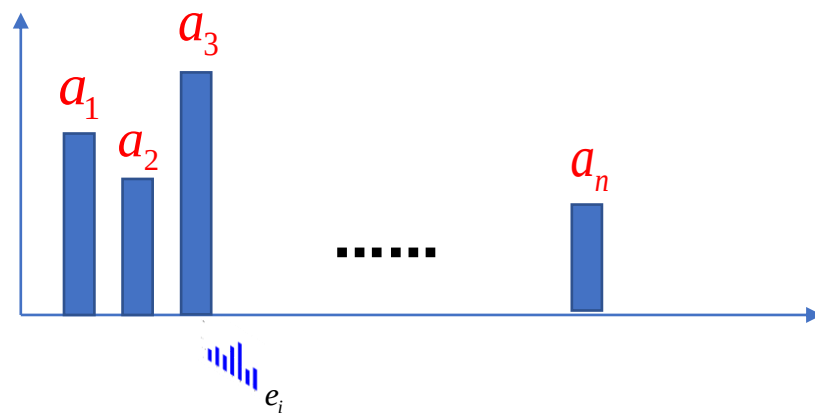
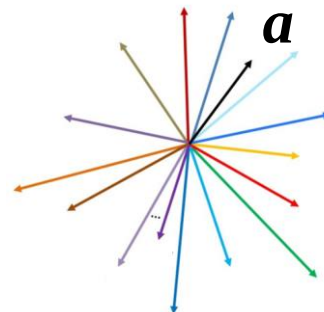
n 维欧式空间

➤ 向量的正交分解

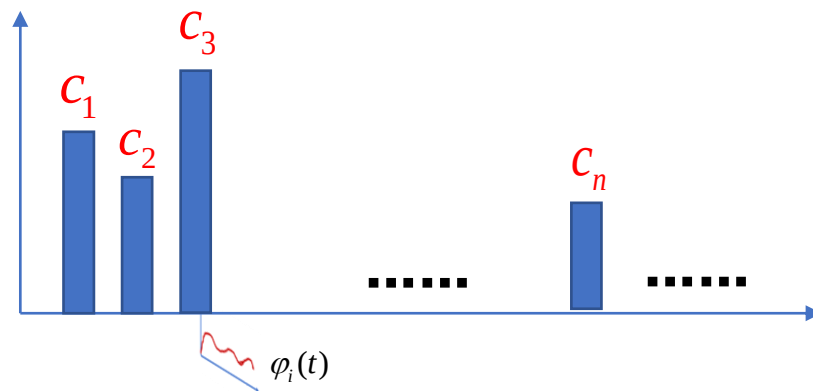
$$\mathbf{a} = \sum_{i=1}^n a_i \mathbf{e}_i$$

$$a_i = \mathbf{a}^T \mathbf{e}_i$$

$$\mathbf{e}_i^T \mathbf{e}_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$



↓ $n \rightarrow \infty$



无限维空间

$$\sum_{i=1}^{\infty} c_i \varphi_i(t)$$

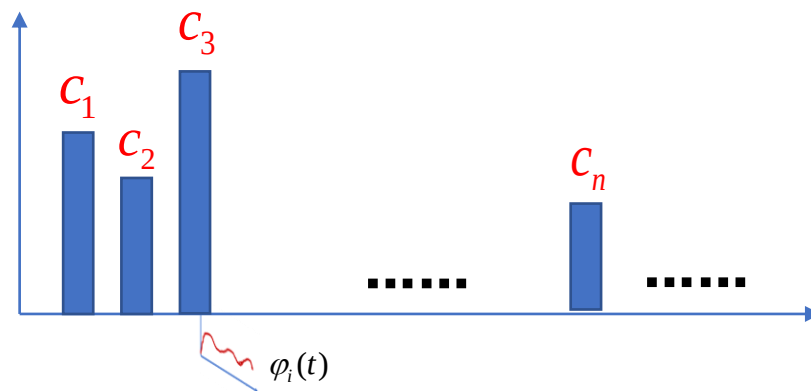
无限维空间



David Hilbert
(1862-1943)

$$a = \sum_{i=1}^{\infty} a_i e_i$$

↓ $n \rightarrow \infty$

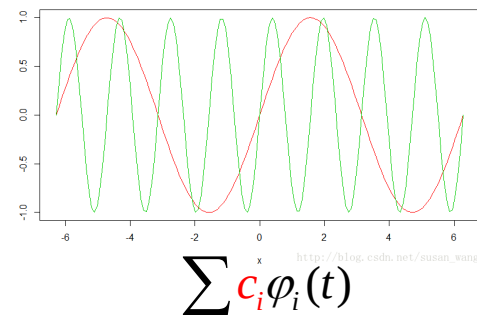
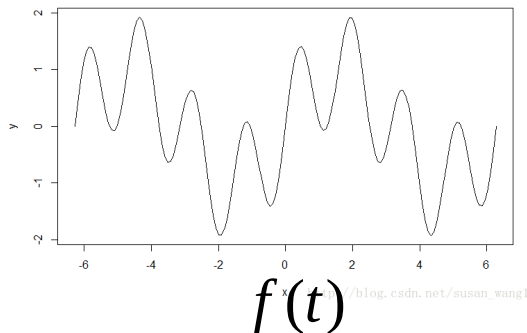


离散的点元素

无限维空间
(希尔伯特空间)

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

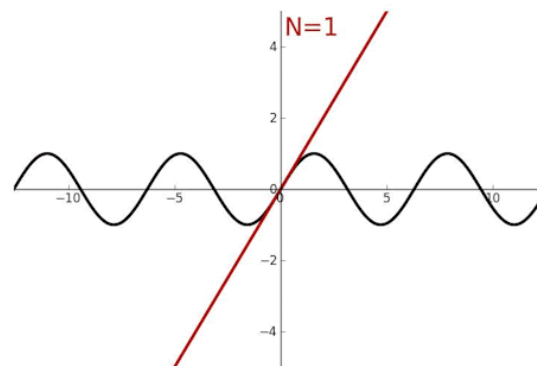
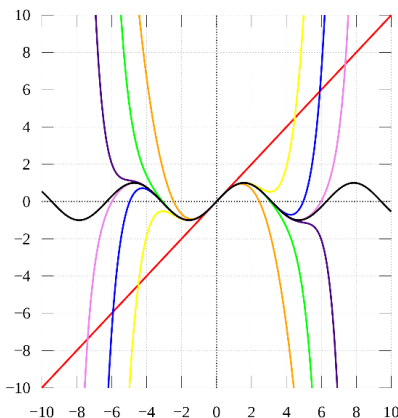
连续函数



无限维空间

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

泰勒公式：无限幂级数展开



$$f(x) = \frac{f(x_0)}{0!} + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n +$$

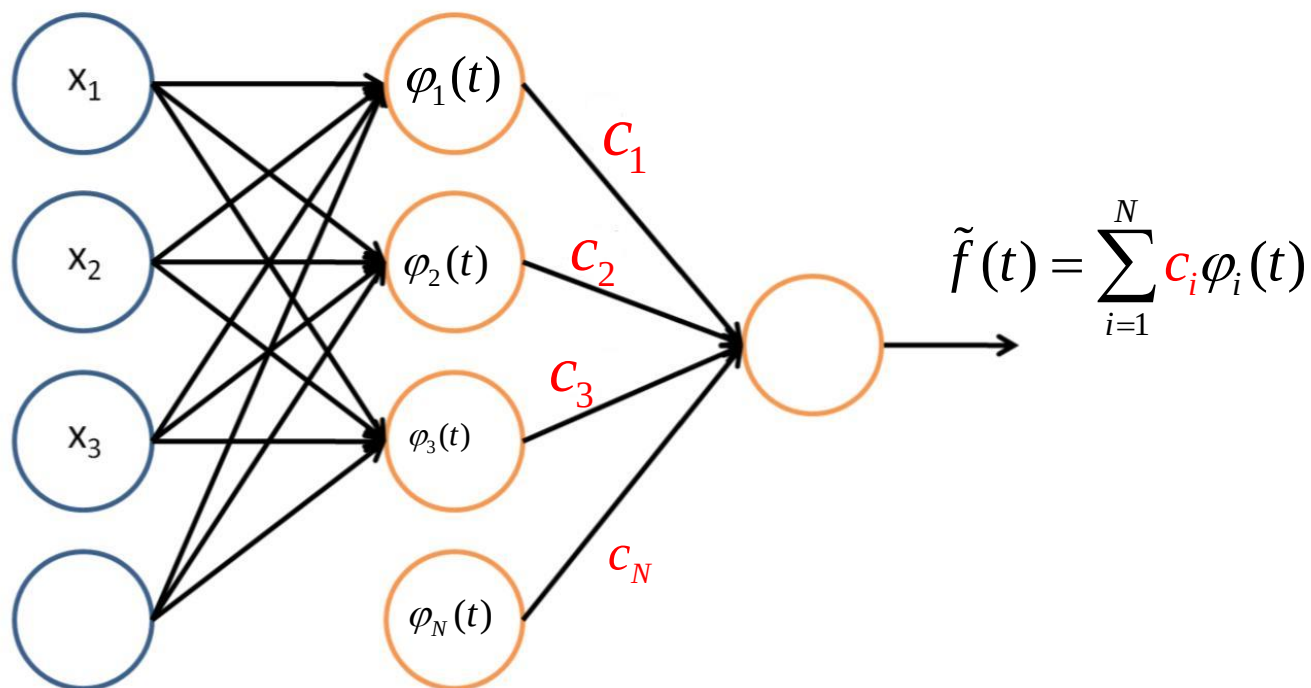
$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x - a)^n$$

函数的正交分解

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无限维空间

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$



无限维空间



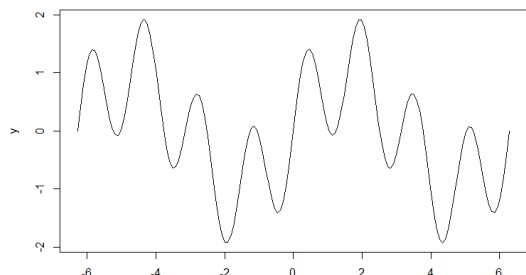
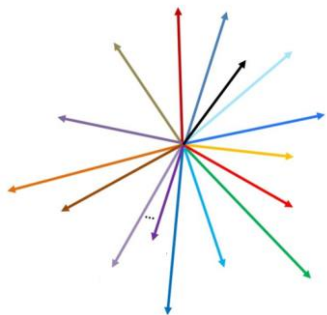
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$$\mathbf{a} = \sum_{i=1}^{\infty} \mathbf{a}_i \mathbf{e}_i$$

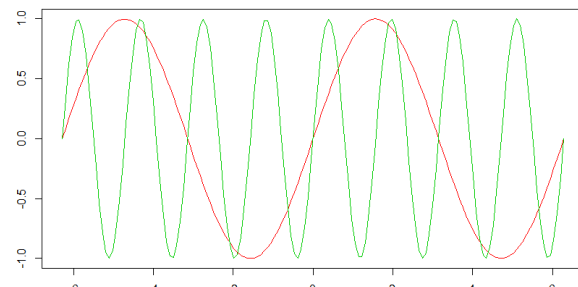
$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

离散的元素

连续函数



http://blog.csdn.net/susan_wang1



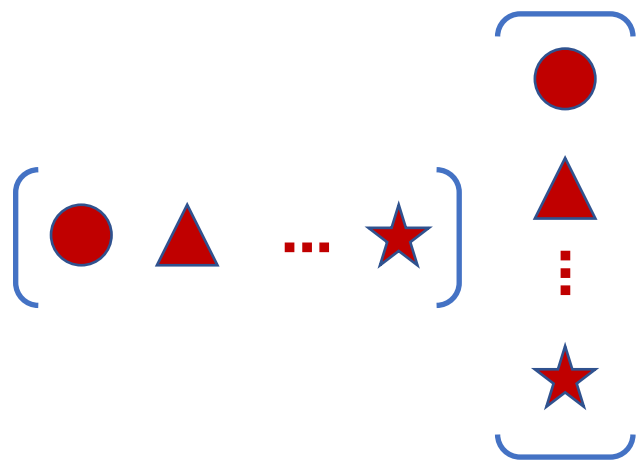
http://blog.csdn.net/susan_wang1

如何界定 $f(t)$ 的模长？如何界定 $\varphi_i(t)$ 为单位函数？
如何界定垂直（正交）？如何界定乘积为0？

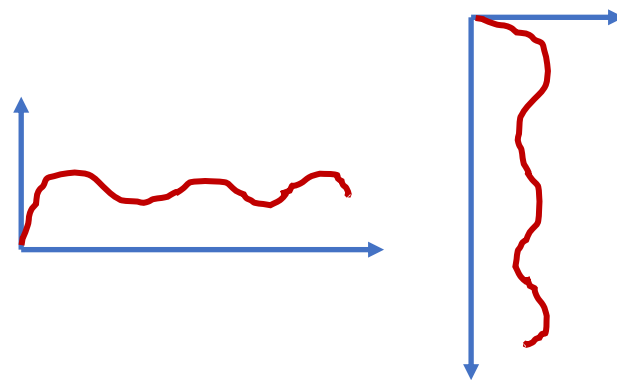
无限维空间

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

如何界定 $f(t)$ 的模长？如何界定 $\varphi_i(t)$ 为单位函数？
如何界定垂直（正交）？如何界定乘积为0？



$$\| \mathbf{a} \|^2 = \mathbf{a}^T \mathbf{a}$$



$$\int_{t_1}^{t_2} \mathbf{f}^T(t) \mathbf{f}(t) dt$$

$$\| \mathbf{f}(t) \|^2$$

$$\mathbf{a} = \sum_{i=1}^{\infty} \mathbf{a}_i \mathbf{e}_i \quad \longrightarrow \quad f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

向量

实函数

复函数

模

$$\sqrt{\mathbf{a}^T \mathbf{a}}$$

$$\sqrt{\int_{t_1}^{t_2} f^2(t) dt}$$

$$\sqrt{\int_{t_1}^{t_2} f(t) f^*(t) dt}$$

正交
(内积)

$$\mathbf{e}_i^T \mathbf{e}_j = 0$$

$$\int_{t_1}^{t_2} \varphi_i(t) \varphi_j(t) dt = 0$$

$$\int_{t_1}^{t_2} \varphi_i(t) \varphi_j^*(t) dt = 0$$

平方可积实函数集合

$$L^2(R) = \left\{ f(t) \mid \int_{-\infty}^{\infty} f^2(t) dt < \infty \right\}$$

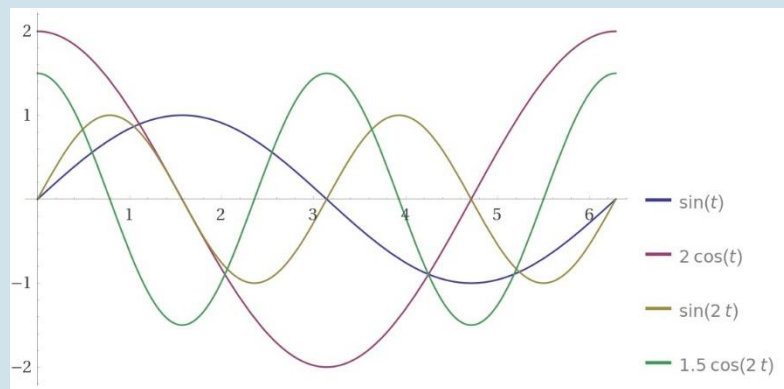
➤ 正交函数集

在 $[t_1, t_2]$ 区间上定义的非零函数序列 $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ ，其中任意两个函数 $\varphi_i(t)$ 与 $\varphi_j(t)$ 均满足条件：

$$\langle \varphi_i, \varphi_j \rangle = \int_{t_1}^{t_2} \varphi_i(t) \varphi_j^*(t) dt = \begin{cases} 0 & i \neq j \\ k_i & i = j \end{cases}$$

其中 k_i 为常数，则称函数序列 $\{\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)\}$ 为在区间 $[t_1, t_2]$ 上的**正交函数集**。

$$\int_{-1}^1 (2t + 3) (45t^2 + 9t - 17) dt = 0$$



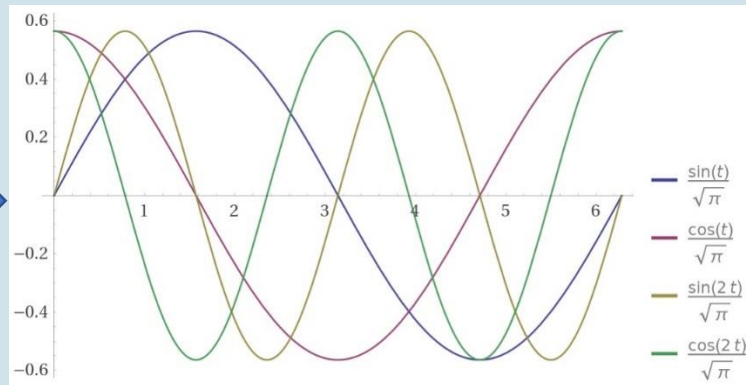
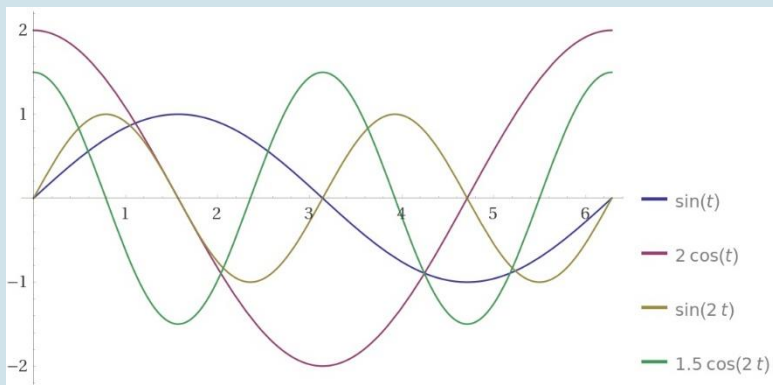
注：*表示共轭。两个实部相等，虚部互为相反数的复数互为共轭复数。

➤ 标准正交函数集

在 $[t_1, t_2]$ 区间上定义的非零函数序列 $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$, 其中任意两个函数 $\varphi_i(t)$ 与 $\varphi_j(t)$ 均满足条件:

$$\langle \varphi_i, \varphi_j \rangle = \int_{t_1}^{t_2} \varphi_i(t) \varphi_j^*(t) dt = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

则称函数序列 $\{\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)\}$ 为在区间 $[t_1, t_2]$ 上的**标准正交函数集**, 可由原正交函数集归一化得到。



➤ 完备的标准正交函数集

如果在 $[t_1, t_2]$ 区间，除正交函数集 $\{\varphi_i(t)\}$ 之外，不存在有非零的函数 $x(t)$ 满足下式：

$$\int_{t_1}^{t_2} x(t) \varphi_i^*(t) dt = 0$$

则称此正交函数集 $\{\varphi_i(t)\}$ 为**完备的正交函数集**。

三角函数集： $\{\sin n\omega_1 t, \cos n\omega_1 t, 1; n = 1, 2, \dots\}$

函数的正交分解

34

当函数 $f(t)$ 在 $[t_1, t_2]$ 区间具有连续的一阶导数和**逐段**连续的二阶导数时, $f(t)$ 可以用完备的正交函数集 $\{\varphi_i(t)\}$ 来表示, 即:

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

其中 c_i 为常数。则称此表示为**函数的正交分解**。

$$\langle \varphi_i(t), \varphi_j(t) \rangle = 0 \quad \Rightarrow \quad f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$\varphi_j(t)$

$$c_j = \frac{\langle f(t), \varphi_j(t) \rangle}{\langle \varphi_j(t), \varphi_j(t) \rangle} = \frac{1}{k_j} \langle f(t), \varphi_j(t) \rangle$$

$$\langle f(t), \varphi_j(t) \rangle = \sum_{i=1}^{\infty} c_i \langle \varphi_i(t), \varphi_j(t) \rangle \quad \Rightarrow \quad \langle f(t), \varphi_j(t) \rangle = c_j \langle \varphi_j(t), \varphi_j(t) \rangle$$

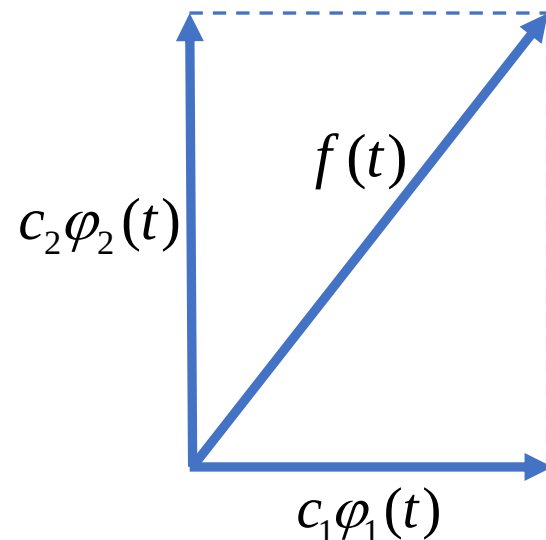
帕斯瓦尔定理：
$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$



Marc-Antoine Parseval
(1755-1836)

- 表示可积函数与其正交分解系数之间关系的恒等式。
 - 函数平方的和等于其正交变换分解系数的平方之和
- 从几何观点来看，这就是内积空间上的毕达哥拉斯定理。
 - 向量的正交分量的平方和等于向量长度（模）的平方



帕斯瓦尔定理：
$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$



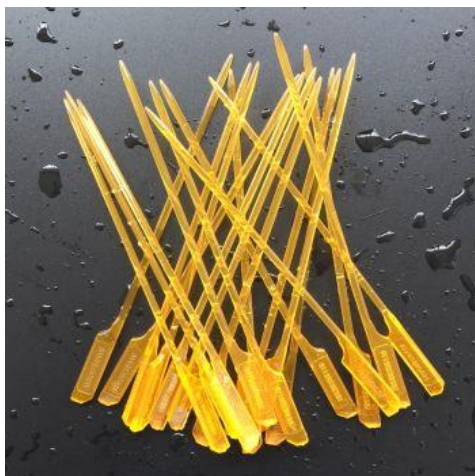
Marc-Antoine Parseval
(1755-1836)

$$\begin{aligned} \int_{t_1}^{t_2} \|f(t)\|^2 dt &= \int_{t_1}^{t_2} \left\| \sum_{i=1}^{\infty} c_i \varphi_i(t) \right\|^2 dt \\ &= \int_{t_1}^{t_2} \left(\sum_{i=1}^{\infty} c_i \varphi_i(t) \right) \left(\sum_{i=1}^{\infty} c_i^* \varphi_i^*(t) \right) dt \\ &= \int_{t_1}^{t_2} \left(\sum_{i=1}^{\infty} \|c_i\|^2 \varphi_i(t) \varphi_i^*(t) + \sum_{i \neq j} c_i c_j^* \varphi_i(t) \varphi_j^*(t) \right) dt \\ &= \sum_{i=1}^{\infty} \|c_i\|^2 k_i \end{aligned}$$

帕斯瓦尔定理：
$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$

信号的能量等于信号在完备正交函数集中各分量的能量之和

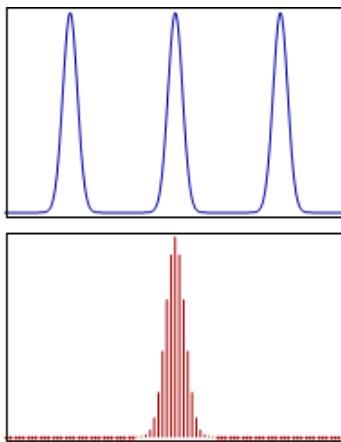


$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

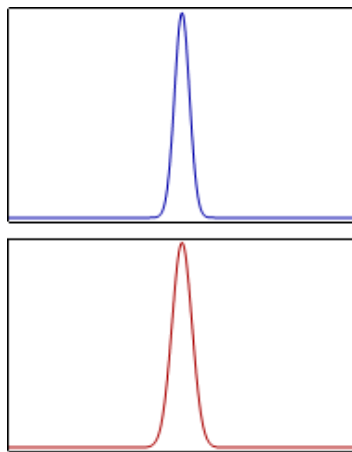
Why?



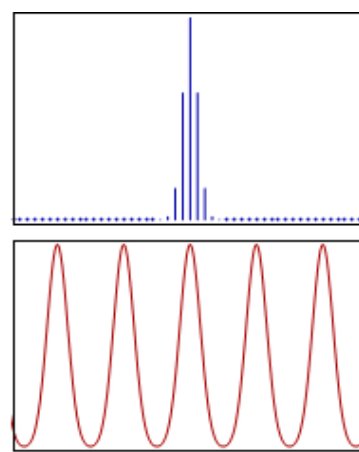
连续周期信号



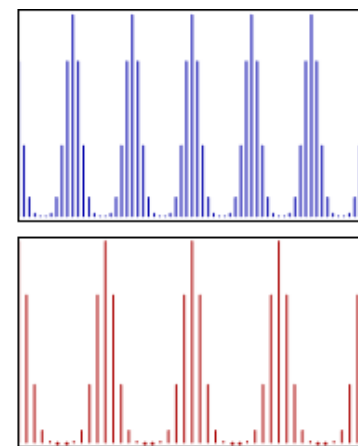
连续信号



离散信号

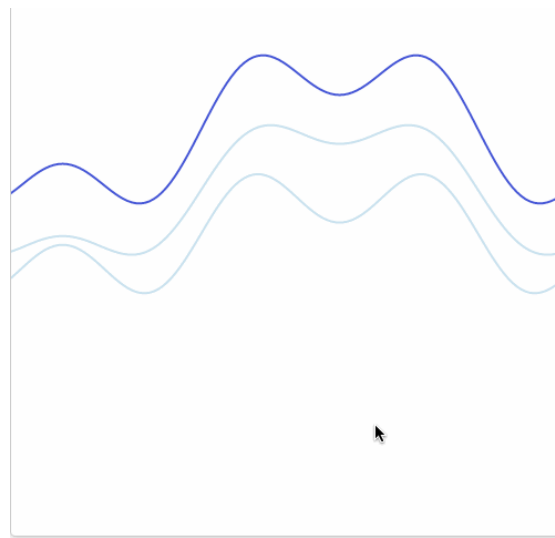
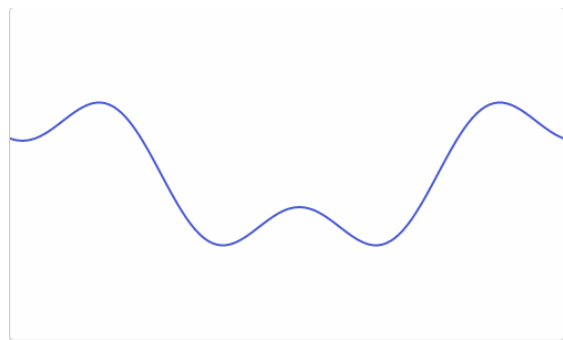


离散周期信号

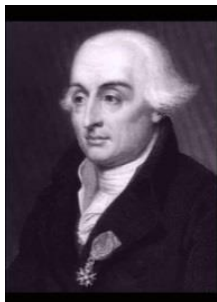


周期信号的傅立叶级数展开

$$f(t) = \sum_{n=1}^{\infty} c_n \varphi_n(t)$$

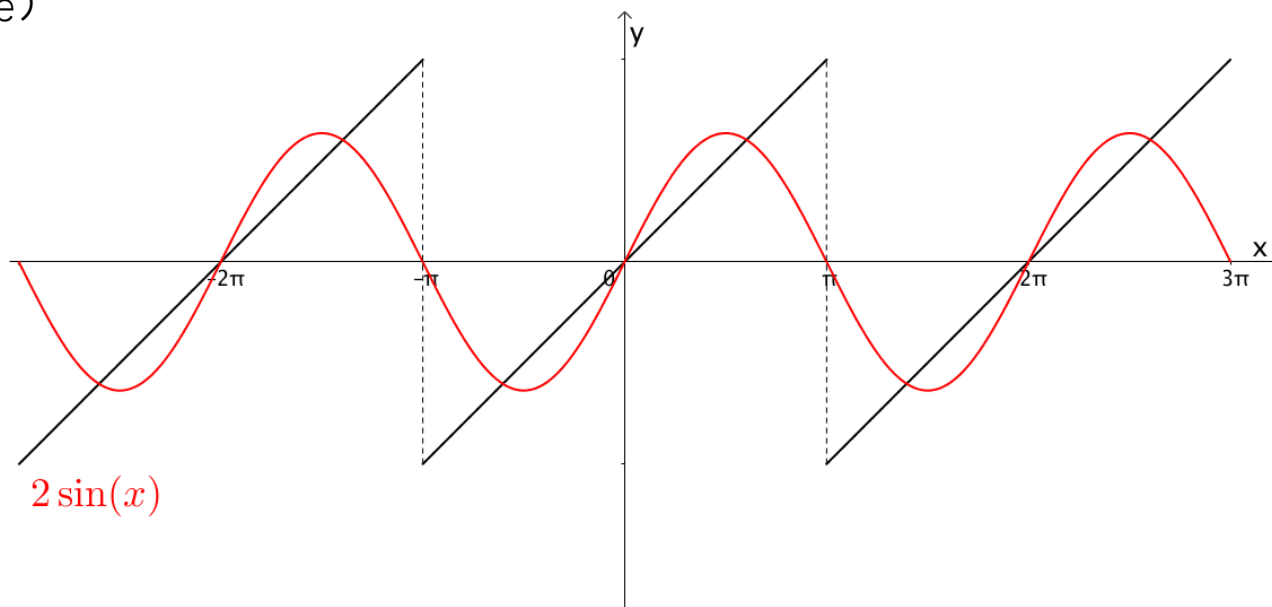


三角函数集： $\{\sin n\omega_1 t, \cos n\omega_1 t, 1; n = 1, 2, \dots\}$



拉格朗日
(Lagrange)

拉格朗日等数学家发现某些周期函数可以由三角函数的和来表示



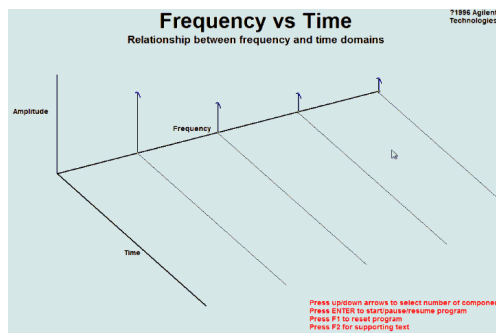
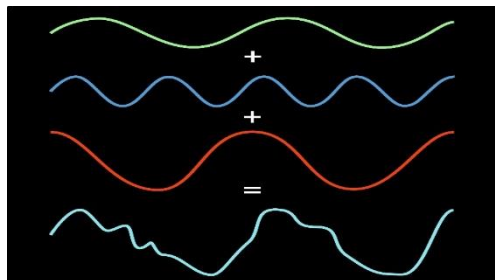
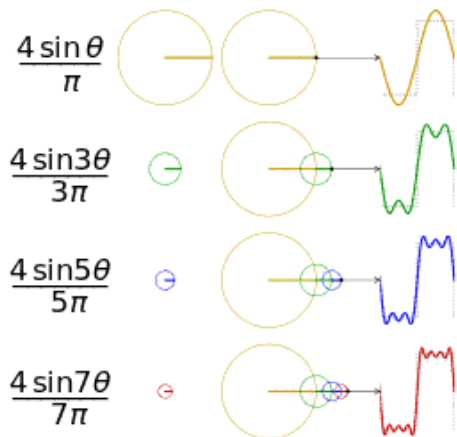
傅立叶级数——概述

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傅立叶
(1768—1830)

任何周期信号都可以表示为不同频率的正弦波信号的无限叠加。

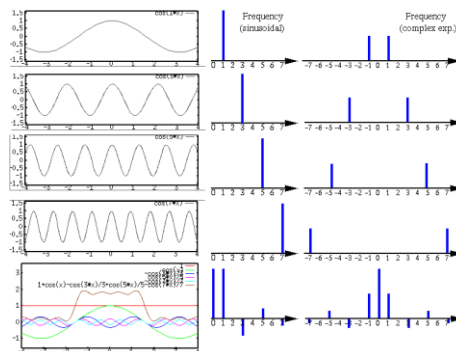
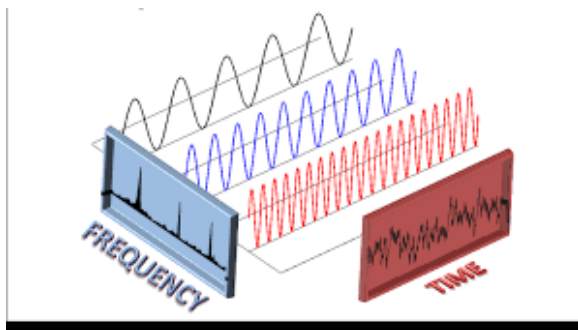


- 将复杂的周期运动转变为无穷多个简谐运动的叠加
- 我们所观察到的周期运动大多数类似于左图中红点的运动。它初看是杂乱无章的，但实际上可以看作四个匀速圆周运动的叠加

傅立叶级数——概述

44

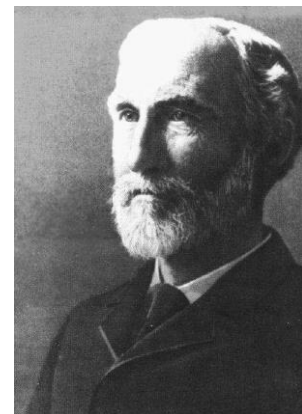
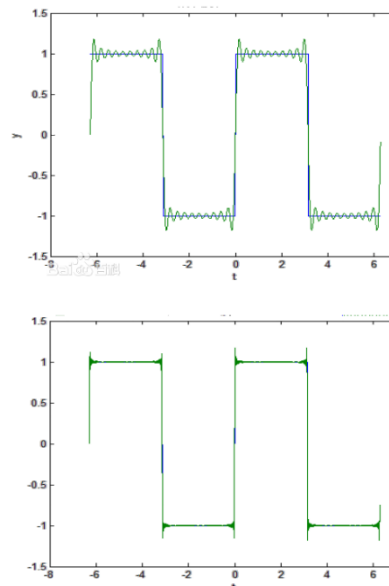
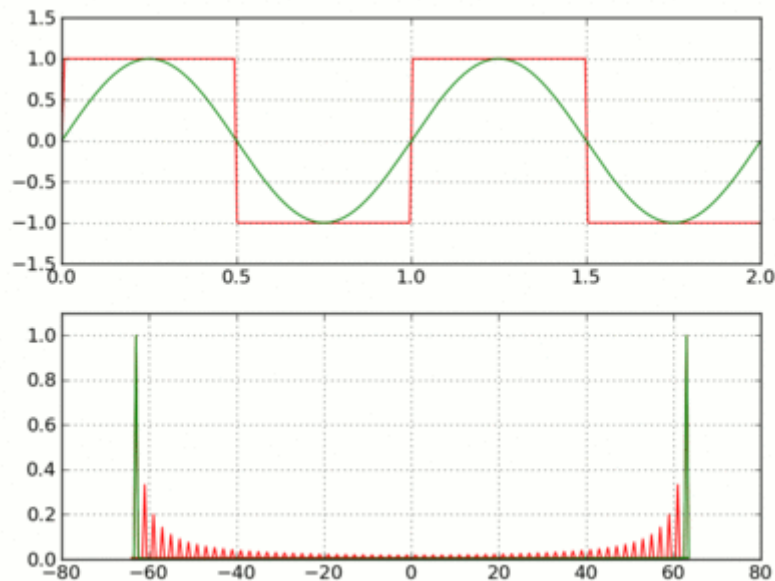
任何周期信号都可以表示为不同频率的正弦波信号的无限叠加。



拉格朗日 (Lagrange)



拉普拉斯 (Laplace)

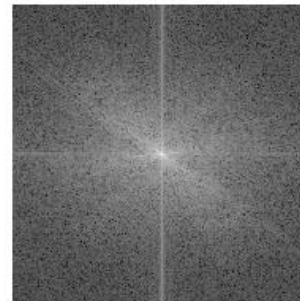
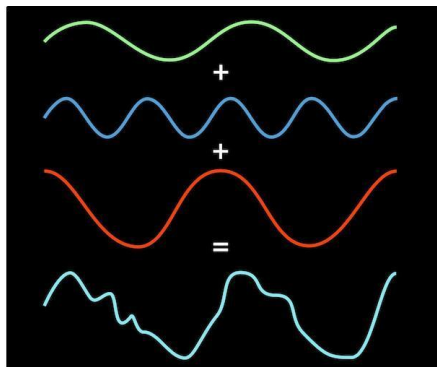


1839-1903

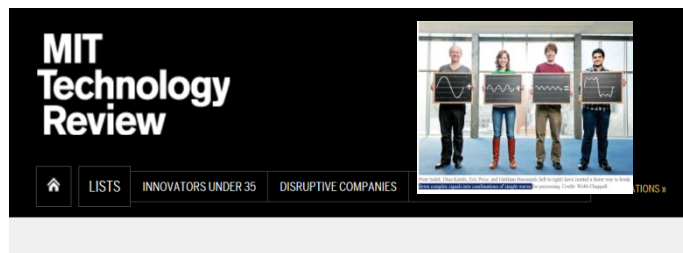
吉布斯现象：在工程应用时常用有限正弦项正弦波叠加逼近原周期信号。所用的谐波次数 N 的大小决定逼近原波形的程度， N 增加，逼近的精度不断改善。但是由于对于具有不连续点的周期信号会发生一种现象：当选取的傅里叶级数的项数 N 增加时，合成的波形虽然更逼近原函数，但在不连续点附近会出现一个固定高度的过冲， N 越大，过冲的最大值越靠近不连续点，但其峰值并不下降，而是大约等于原函数在不连续点处跳变值的9%，且在不连续点两侧呈现衰减振荡的形式。

$$\text{Si}(\pi)/\pi - 1/2 \approx 8.95\%.$$

$$\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt,$$



应用：Jpeg, MP3, Wi-Fi路由器, 4G无线通信网络……



IO BREAKTHROUGH
TECHNOLOGIES

2012

A Faster Fourier Transform

A mathematical upgrade promises a speedier digital world

2012年10项改变世界的新技术---稀疏傅里叶变换新算法

美国《技术评论》评出2012年10项改变世界的新技术---稀疏傅里叶变换新算法榜上有名

今年1月，4名麻省理工学院研究人员向人们展示了一种计算机科学中最重要算法的替代品。迪娜·凯塔比、海赛姆·哈桑尼、彼得·因迪克、埃里克·普赖斯创建出一种进行傅里叶变换的更快方法。傅里叶变换是一种处理数据流的数学技术，其已成为Wi-Fi路由器、数字医疗影像和4G蜂窝网络的运行基础。

傅里叶变换的历史可追溯到19世纪，其原理是，任何信号，如录音信号，可表示为不同频率和振幅的正弦和余弦波的集合总和。对波的集合处理起来相对容易，例如，可对录音信号进行压缩或对噪音进行抑制。在20世纪60年代中期，又发展出了一种被称为快速傅里叶变换（FFT）的对计算机友好的算法。快速傅里叶变换的威力到底有多大，你只需比较一下MP3文件及其未压缩格式的大小即可。

➤ 傅立叶级数展开

- 如果正交函数集是**三角函数集**，则周期函数展成的级数就是“**傅立叶级数**”。
- 相应的级数通常被称为“**三角形形式傅立叶级数**”。

$$f(t) = \sum_{n=1}^{\infty} c_n \varphi_n(t)$$

$$\text{三角函数集} \quad \{1, \cos n\omega_1 t, \sin n\omega_1 t\} \quad n \in N$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

➤ 狄义赫利条件

- 满足狄义赫利条件的周期函数都可以在一组正交基函数上展开成为无穷级数。
- 狄义赫利条件：在一个周期内
 - (1) 间断点的个数有限
 - (2) 极值点的个数有限
 - (3) 绝对积分数值有限

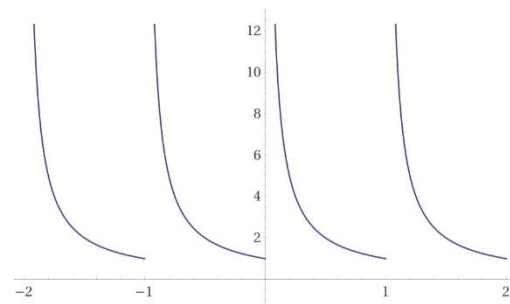
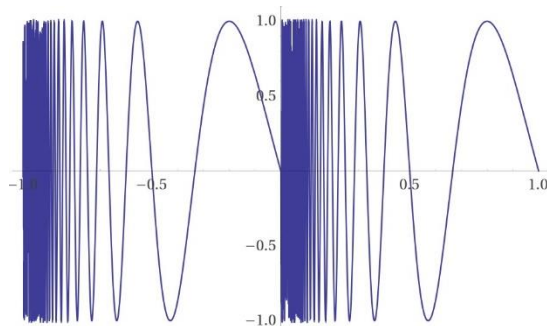
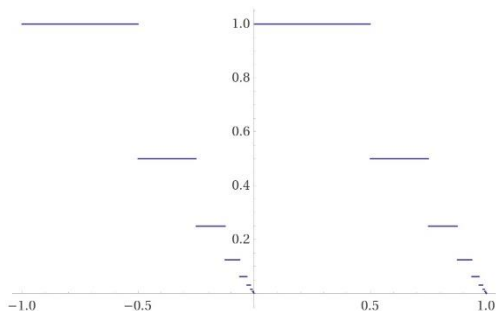


Johann Peter Gustav
Lejeune Dirichlet (1805-
1859)

$$\int_{t_0}^{t_0+T} |f(t)| dt < \infty$$

狄义赫利条件 反例

- (1) 间断点的个数有限 (2) 极值点的个数有限 (3) 绝对可积



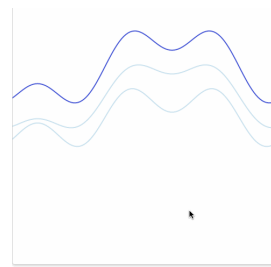
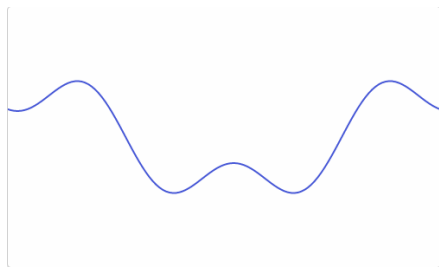
信号的周期为1，后一个阶梯的高度和宽度都是前一个阶梯的一半，可见一个周期内的面积不会超过1，但间断点的数目是无穷多个。

$$f(t) = \sin\left(\frac{2\pi}{t}\right), 0 < t < 1 \quad f(t) = \frac{1}{t}, 0 < t < 1$$

➤ 三角形式傅立叶级数

- 设周期函数 $f(t)$ 的周期为 T_1
- 函数 $\{1, \cos(n\omega_1 t), \sin(n\omega_1 t)\}$ 是正交函数集, 令 $T_1 = 2\pi / \omega_1$
- 则展开成三角函数的无穷级数形式:

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

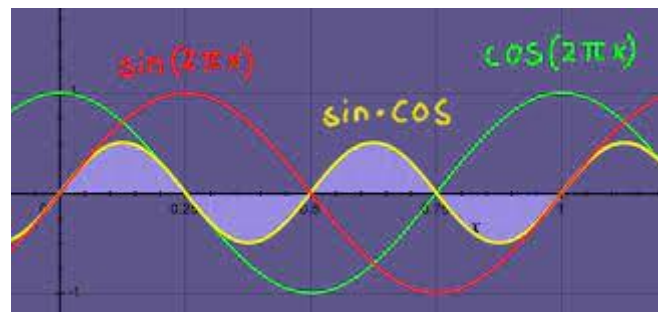


➤ 三角形式傅立叶级数

- 函数 $\{1, \cos(n\omega_1 t), \sin(n\omega_1 t)\}$ 是正交函数集

$$\int_{t_0}^{t_0+T_1} \sin m\omega_1 t \cos n\omega_1 t dt = 0$$

$$\begin{aligned} & \int_{t_0}^{t_0+T_1} \sin m\omega_1 t \cos n\omega_1 t dt \\ &= \int_{t_0}^{t_0+T_1} \frac{1}{2} [\sin(m+n)\omega_1 t + \sin(m-n)\omega_1 t] dt \\ &= 0 \end{aligned}$$



➤ 三角形式傅立叶级数

$$\int_{t_0}^{t_0+T_1} \cos m\omega_1 t \cos n\omega_1 t dt = \begin{cases} \frac{T_1}{2} & m = n \neq 0 \\ T_1 & m = n = 0 \\ 0 & m \neq n \end{cases}$$

$$\int_{t_0}^{t_0+T_1} \sin m\omega_1 t \sin n\omega_1 t dt = \begin{cases} \frac{T_1}{2} & m = n \neq 0 \\ 0 & \text{others} \end{cases}$$

$$\begin{aligned} & \int_{t_0}^{t_0+T_1} \cos m\omega_1 t \cos n\omega_1 t dt \\ &= \int_{t_0}^{t_0+T_1} \frac{1}{2} [\cos(m-n)\omega_1 t + \cos(m+n)\omega_1 t] dt \\ &= \begin{cases} \frac{T_1}{2} & m = n \neq 0 \\ T_1 & m = n = 0 \\ 0 & m \neq n \end{cases} \end{aligned}$$

$$\begin{aligned} & \int_{t_0}^{t_0+T_1} \sin m\omega_1 t \sin n\omega_1 t dt \\ &= \int_{t_0}^{t_0+T_1} \frac{1}{2} [\cos(m-n)\omega_1 t - \cos(m+n)\omega_1 t] dt \\ &= \begin{cases} \frac{T_1}{2} & m = n \neq 0 \\ 0 & \text{others} \end{cases} \end{aligned}$$

➤ 三角形式傅立叶级数

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

- 系数 a_n 和 b_n 统称为三角形式的傅里叶级数系数，简称为**傅立叶系数**。系数计算：

$$c_n = \frac{\langle f(t), \varphi_n(t) \rangle}{\langle \varphi_n(t), \varphi_n(t) \rangle} = \frac{\langle f(t), \varphi_n(t) \rangle}{k_n} = \frac{1}{k_n} \int_{t_1}^{t_2} f(t) \varphi_n^*(t) dt$$

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt$$

直流分量的振幅

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos n\omega_1 t dt$$

余弦分量的振幅

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin n\omega_1 t dt$$

正弦分量的振幅

➤ 三角形式傅立叶级数

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

紧凑型三角形式FS:

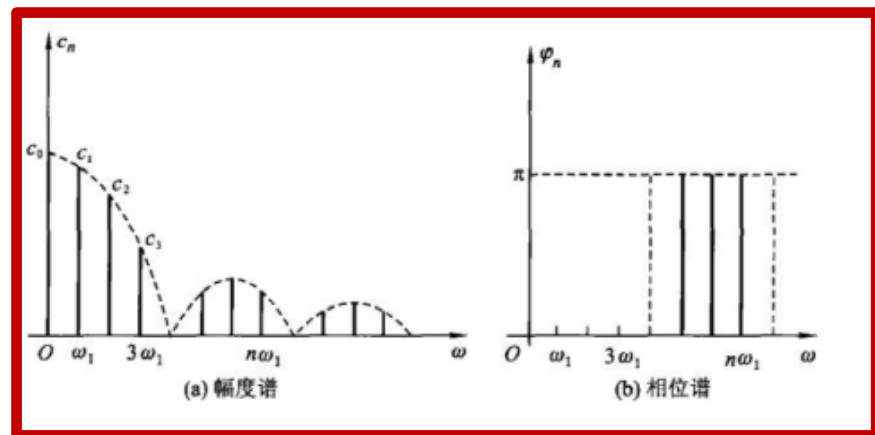
$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n)$$

$$\left\{ \begin{array}{l} a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt \\ a_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos n\omega_1 t dt \\ b_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin n\omega_1 t dt \end{array} \right. \quad \longrightarrow \quad \left\{ \begin{array}{l} c_0 = a_0 \\ c_n = \sqrt{a_n^2 + b_n^2} \\ \tan \varphi_n = -\frac{b_n}{a_n} \end{array} \right.$$

➤ 三角形式傅立叶级数

紧凑型三角形式FS: $f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n)$

其中
$$\begin{cases} c_0 = a_0 \\ c_n = \sqrt{a_n^2 + b_n^2} \\ \tan \varphi_n = -\frac{b_n}{a_n} \end{cases}$$

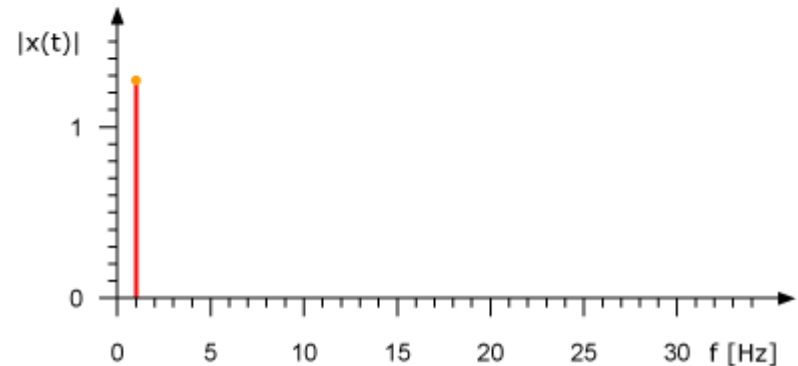
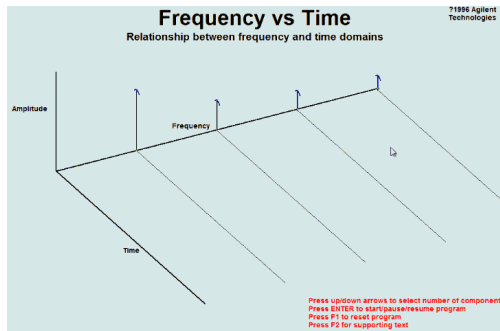
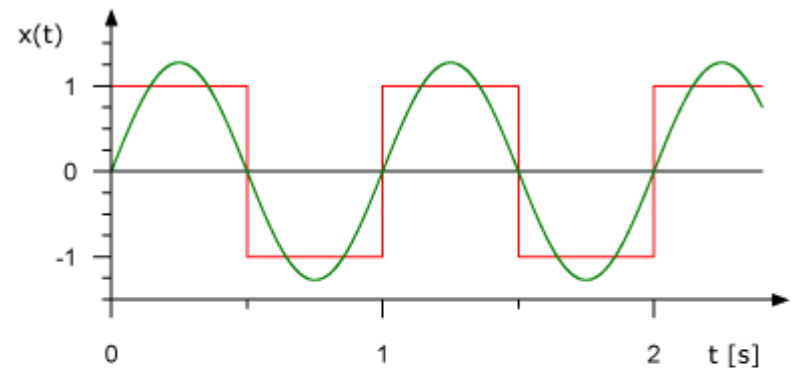
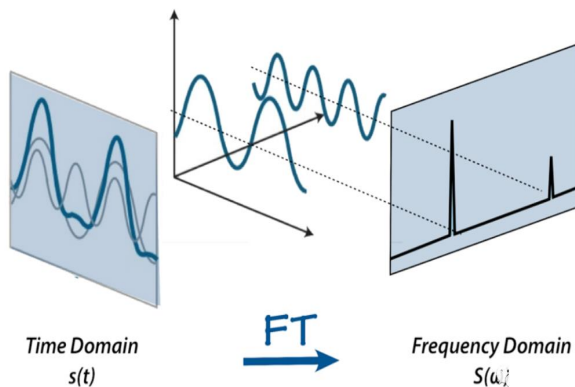


c_n, φ_n 是频率的函数，其图形称为频谱图

$c_n \sim \omega = n\omega_1$ 的关系图称为**振幅谱**

$\varphi_n \sim \omega = n\omega_1$ 的关系图称为**相位谱**

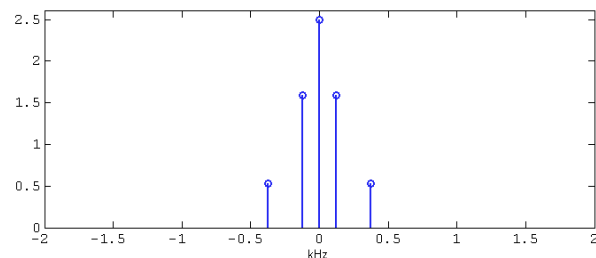
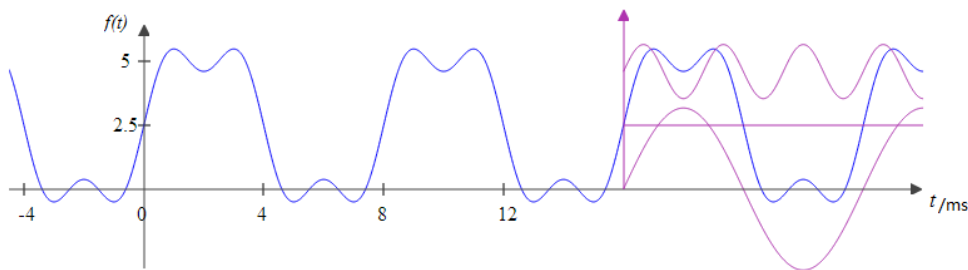
$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n)$$



➤ 三角形式傅立叶级数

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \\ &= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \end{aligned}$$
$$\begin{cases} a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt \\ a_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos n\omega_1 t dt \\ b_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin n\omega_1 t dt \end{cases} \quad \begin{cases} c_n = \sqrt{a_n^2 + b_n^2} \\ \tan \varphi_n = -\frac{b_n}{a_n} \end{cases}$$

任何周期信号只要满足Dirichlet条件，就可以分解成直流分量和许多简谐振荡分量的叠加



➤ 三角形式傅立叶级数

特例1：偶信号  余弦级数

$$f(-t) = f(t)$$

$$a_0 = \frac{2}{T} \int_0^{T/2} f(t) dt$$

$$a_n = \frac{4}{T} \int_0^{T/2} f(t) \cos n\omega_1 t dt$$

$$b_n = 0$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_1 t$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos n\omega_1 t dt$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin n\omega_1 t dt$$

➤ 三角形式傅立叶级数

特例2：奇信号  正弦级数

$$f(-t) = -f(t)$$

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{4}{T} \int_0^{T/2} f(t) \sin n\omega_1 t dt$$

$$f(t) = \sum_{n=1}^{\infty} b_n \sin n\omega_1 t$$

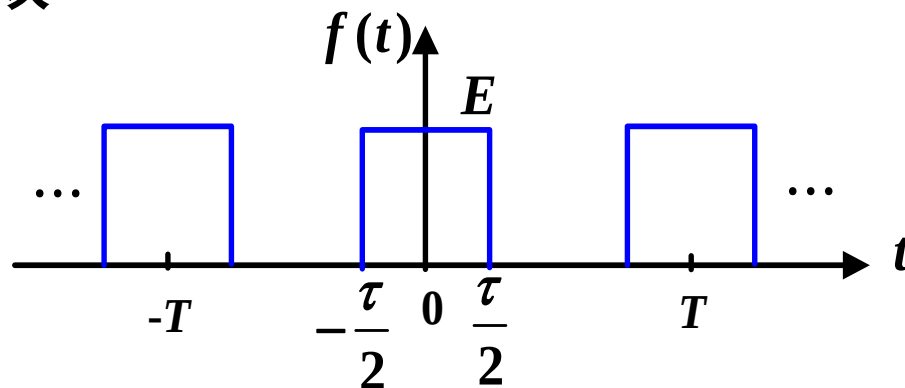
$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos n\omega_1 t dt$$

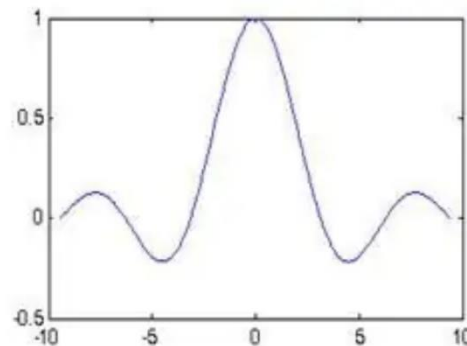
$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin n\omega_1 t dt$$

➤ 三角形式傅立叶级数

例1 求图所示周期为 T 的矩形脉冲信号的FS和频谱图。

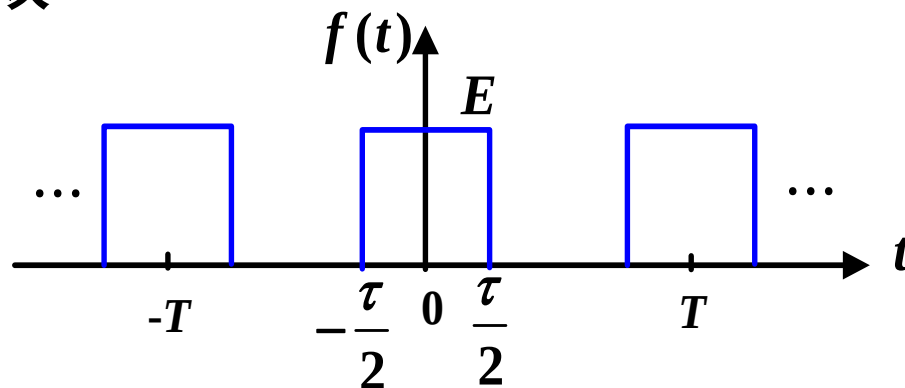


$$\begin{aligned} a_n &= \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos n\omega_1 t dt = \frac{2E}{T} \int_{-\tau/2}^{\tau/2} \cos n\omega_1 t dt = \frac{2E}{T} \frac{1}{n\omega_1} \sin n\omega_1 t \Big|_{-\tau/2}^{\tau/2} = \frac{2E}{T} \frac{1}{n\omega_1} \cdot 2 \sin\left(\frac{n\omega_1 \tau}{2}\right) \\ &= \frac{2E}{n\pi} \sin\left(\frac{n\omega_1 \tau}{2}\right) = \frac{2E}{n\pi} \frac{\sin\left(\frac{n\omega_1 \tau}{2}\right)}{\frac{n\omega_1 \tau}{2}} \frac{n\omega_1 \tau}{2} \\ &= \frac{2E\tau}{T} \text{Sa}\left(\frac{n\omega_1 \tau}{2}\right) \end{aligned}$$



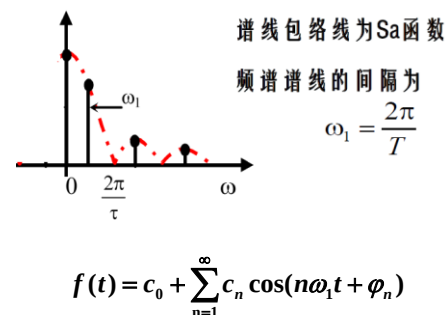
➤ 三角形式傅立叶级数

例1 求图所示周期为 T 的矩形脉冲信号的FS和频谱图。



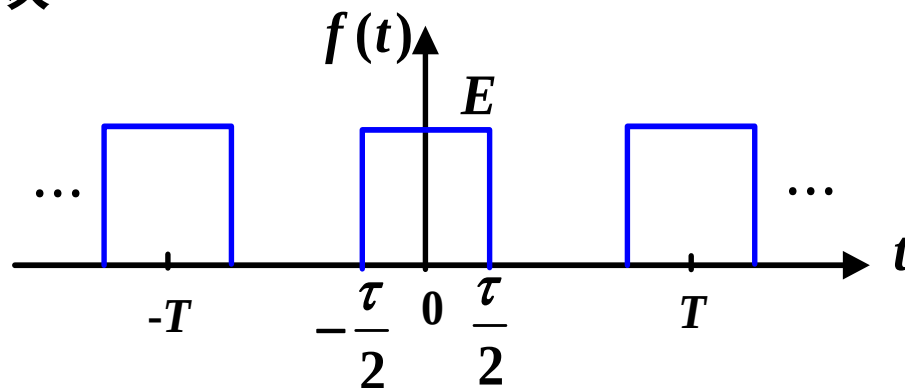
解: $c_0 = a_0 = \frac{E\tau}{T}$

$$c_n = \sqrt{a_n^2 + b_n^2} = |a_n| = \left| \frac{2E\tau}{T} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \right|$$



➤ 三角形式傅立叶级数

例1 求图所示周期为 T 的矩形脉冲信号的FS和频谱图。

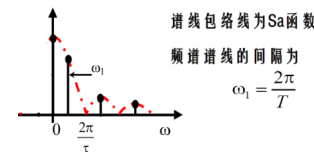


解::

$$c_0 = a_0 = \frac{E\tau}{T}$$

$$c_n = \sqrt{a_n^2 + b_n^2} = |a_n| = \left| \frac{2E\tau}{T} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \right|$$

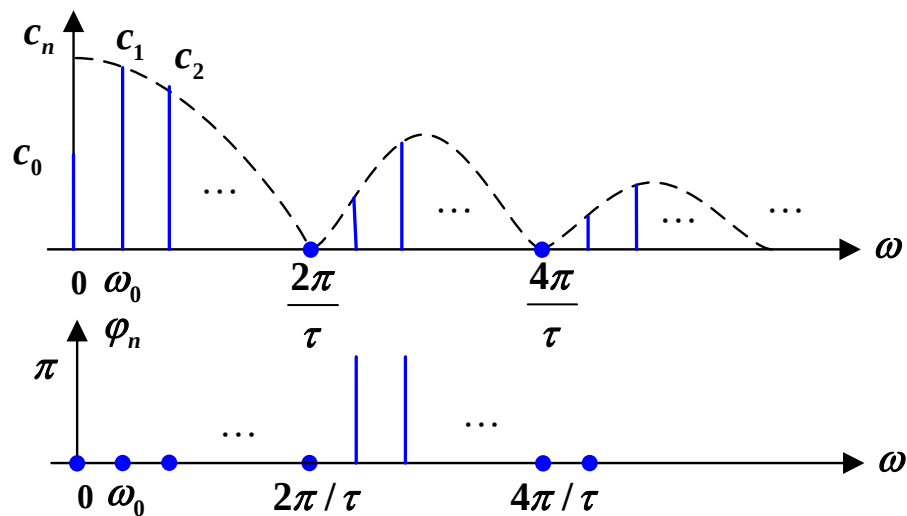
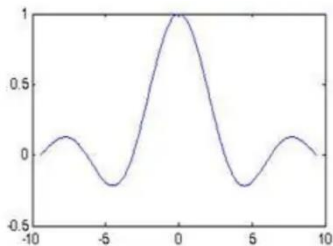
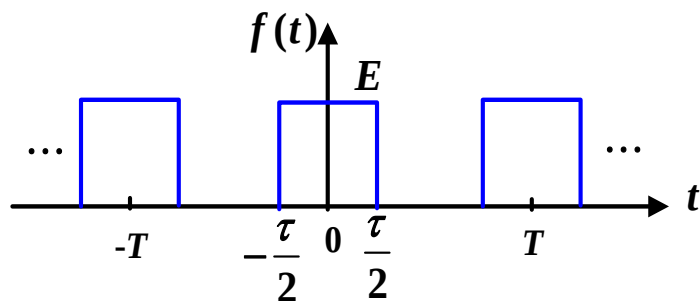
$$\tan \varphi_n = -\frac{b_n}{a_n} = 0 \quad \longrightarrow \quad \varphi_n = \begin{cases} 0 & a_n > 0 \\ \pi & a_n < 0 \end{cases}$$



$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n)$$

➤ 三角形式傅立叶级数

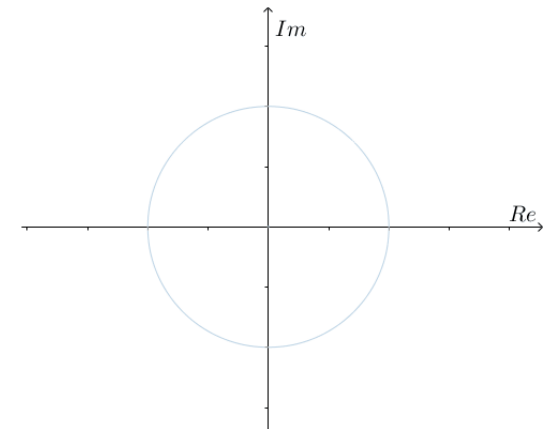
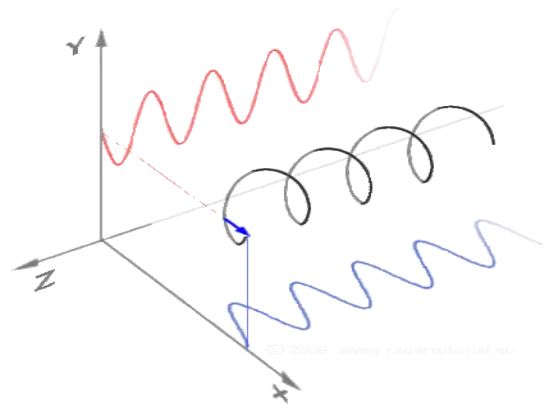
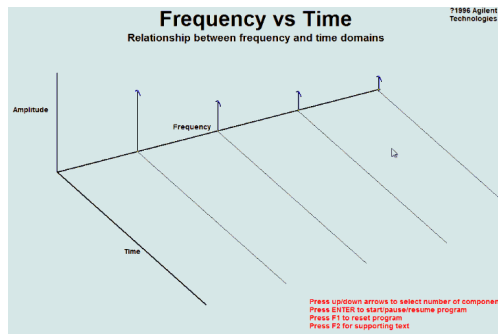
$$c_0 = \frac{E\tau}{T}, \quad c_n = \left| \frac{2E\tau}{T} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \right| \quad \varphi_n = \begin{cases} 0 & a_n > 0 \\ \pi & a_n < 0 \end{cases}$$



周期矩形脉冲信号频谱图

➤ 三角形式傅立叶级数

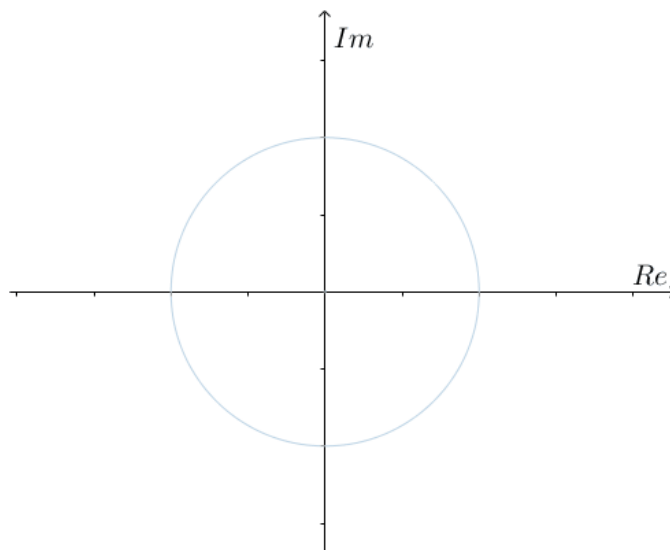
$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



$$e^{j\varphi} = \cos(\varphi) + j \sin(\varphi)$$

欧拉公式

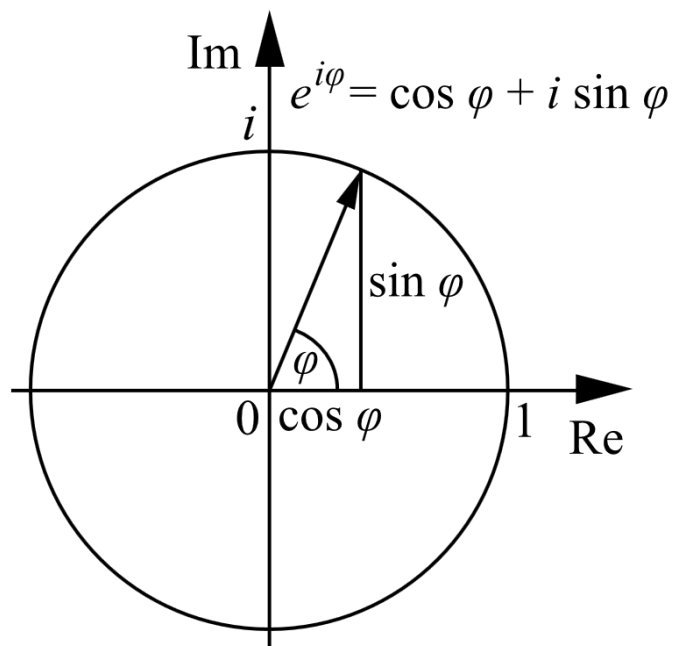
欧拉公式将**三角函数**与**复数指数函数**相关联



$$e^{j\varphi} = \cos(\varphi) + j \sin(\varphi)$$

欧拉公式把实数的**三角**运算变成了复数的**旋转**运算，把**指数**运算变成了**乘积**运算，把**纯微分方程**的求解过程变成了**指数方程**的求解过程，大大简化了运算。

$$e^{j\varphi} = \cos(\varphi) + j \sin(\varphi)$$



$$\sin(\varphi) = \frac{e^{j\varphi} - e^{-j\varphi}}{2j}$$

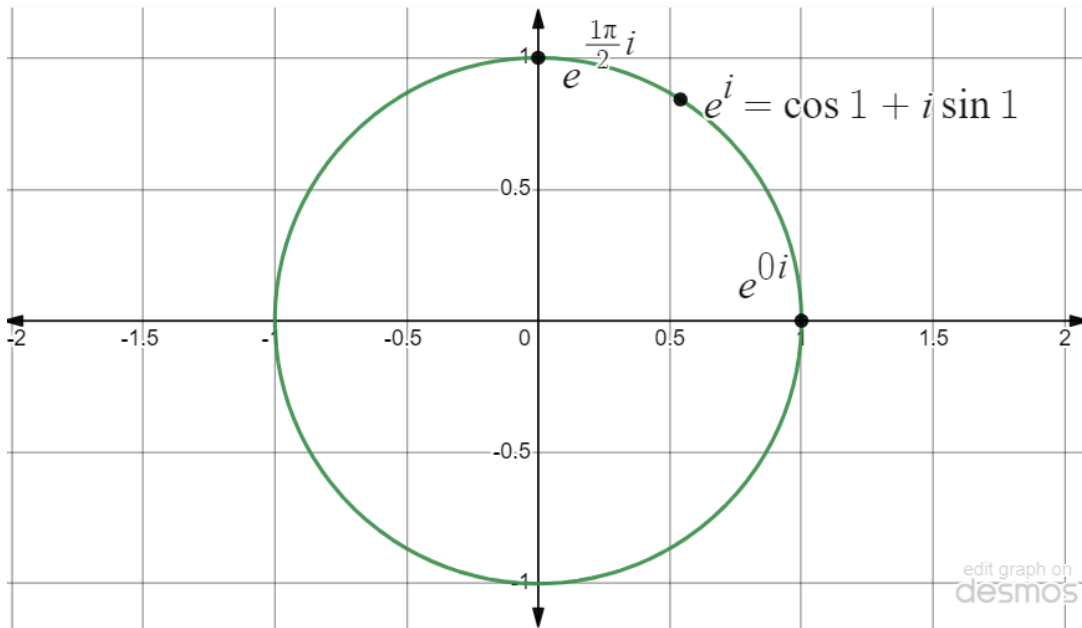
$$\cos(\varphi) = \frac{e^{j\varphi} + e^{-j\varphi}}{2}$$

欧拉公式

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$$e^{j\varphi} = \cos(\varphi) + j \sin(\varphi)$$

$$(e^j)^\varphi = \cos(\varphi) + j \sin(\varphi)$$



$$e^0 = \cos(0) + j \sin(0)$$

$$e^j = \cos(1) + j \sin(1)$$

$$e^{j\frac{\pi}{2}} = \cos\left(\frac{\pi}{2}\right) + j \sin\left(\frac{\pi}{2}\right)$$

$$e^{j\pi} = \cos(\pi) + j \sin(\pi)$$

$$e^{j2\pi} = \cos(2\pi) + j \sin(2\pi)$$

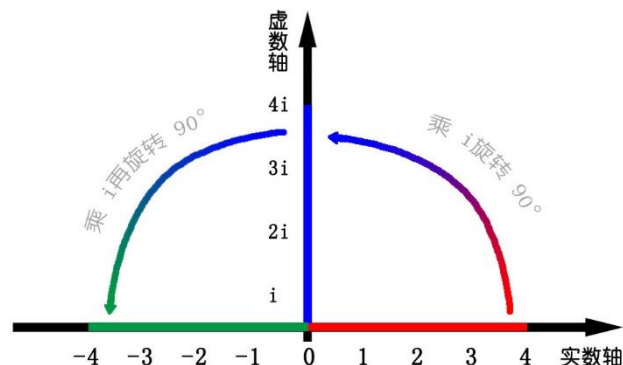
欧拉公式

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$$e^{j\varphi} = \cos(\varphi) + j \sin(\varphi)$$

上帝公式

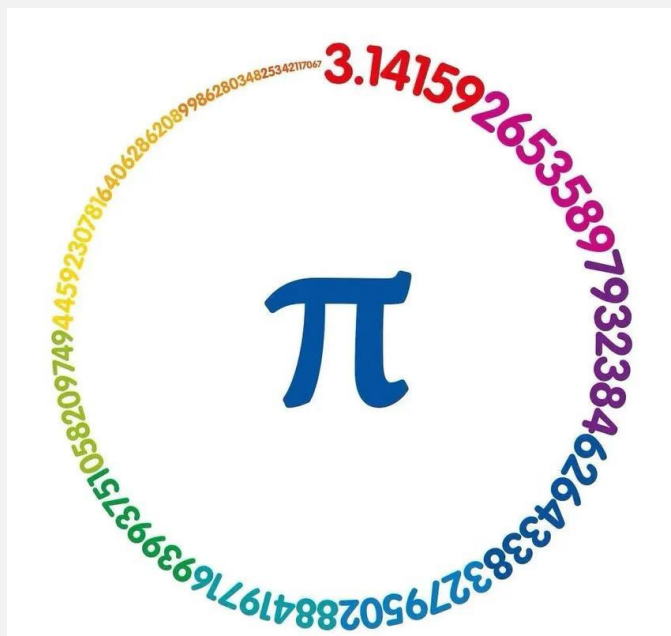
$$e^{j\pi} + 1 = 0$$



- 一个人第一次看到这个公式而不感到它的魅力，他不可能成为数学家。

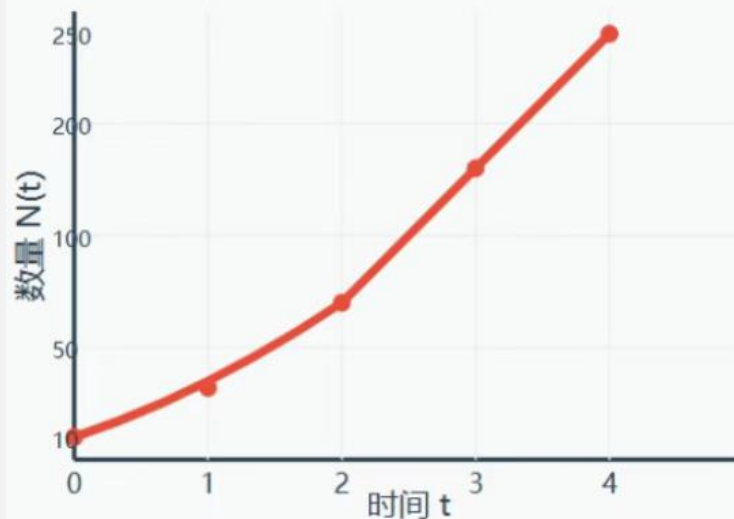
$$e^{j\pi} + 1 = 0$$

π



e

指数增长曲线



➤ 方法一：泰勒级数法

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \frac{(i\theta)^7}{7!} + \frac{(i\theta)^8}{8!} + \dots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \frac{\theta^6}{6!} - \frac{i\theta^7}{7!} + \frac{\theta^8}{8!} + \dots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \frac{\theta^8}{8!} - \dots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right) \\ &= \cos \theta + i \sin \theta \end{aligned}$$

➤ 方法二：微分法

$$f(x) = \frac{\cos x + i \sin x}{e^{ix}}$$

$$f'(x) = \frac{(-\sin x + i \cos x) \cdot e^{ix} - (\cos x + i \sin x) \cdot i \cdot e^{ix}}{(e^{ix})^2}$$

$$= \frac{-\sin x \cdot e^{ix} - i^2 \sin x \cdot e^{ix}}{(e^{ix})^2}$$

$$= \frac{-\sin x \cdot e^{ix} + \sin x \cdot e^{ix}}{(e^{ix})^2}$$

$$= 0$$

➤ 复指数信号

$$f(t) = Ke^{st}$$

复指数信号与正余弦信号之间的关系

$$\begin{aligned} f(t) &= Ke^{st} = Ke^{(\sigma + j\omega)t} \\ &= Ke^{\sigma t} \cdot e^{+j\omega t} \\ &= Ke^{\sigma t} \cdot (\cos \omega t + j \sin \omega t) \end{aligned}$$

欧拉公式

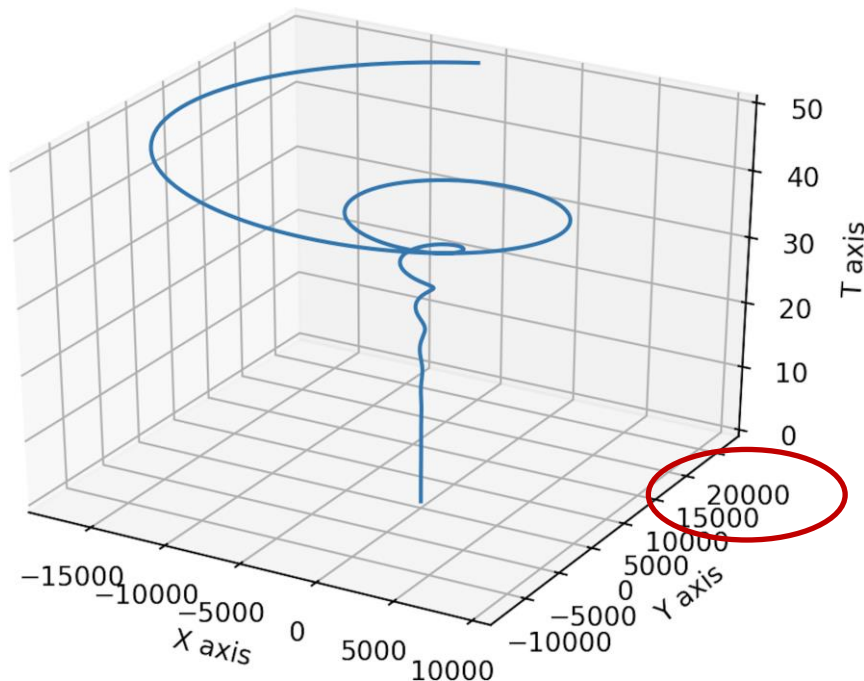


$$\begin{cases} e^{j\omega t} = \cos \omega t + j \sin \omega t \\ e^{-j\omega t} = \cos \omega t - j \sin \omega t \end{cases} \Leftrightarrow \begin{cases} \cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2} \\ \sin \omega t = \frac{e^{j\omega t} - e^{-j\omega t}}{2j} \end{cases}$$

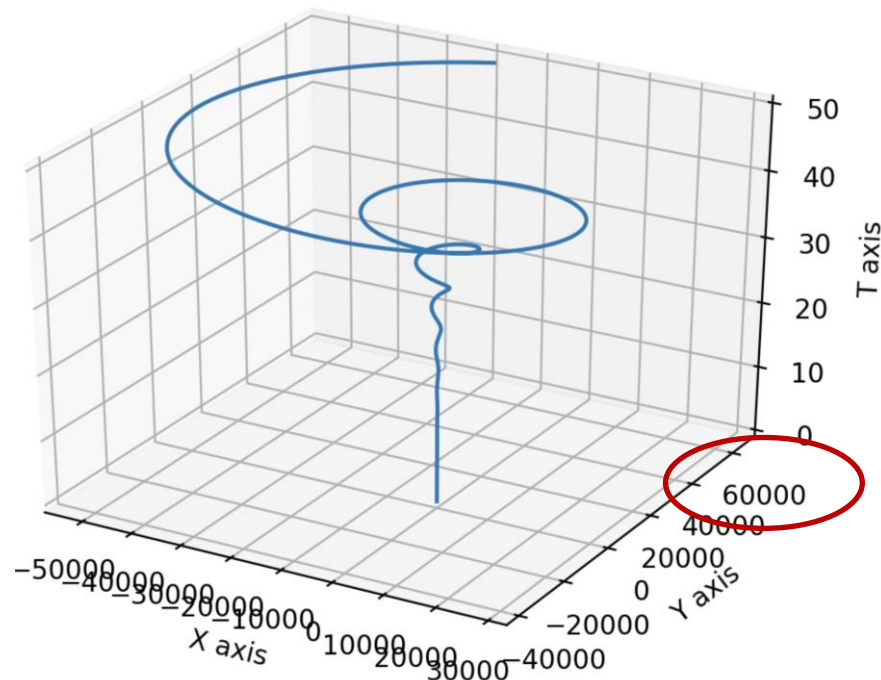
➤ 复指数信号 $f(t) = Ke^{st} = Ke^{\sigma t} \cdot (\cos \omega t + j \sin \omega t)$

K, σ 和 ω 的取值不同，复指数信号有什么不同？

$\sigma = 0.2, \omega = 1, K$ 取不同的值



$K = 1$

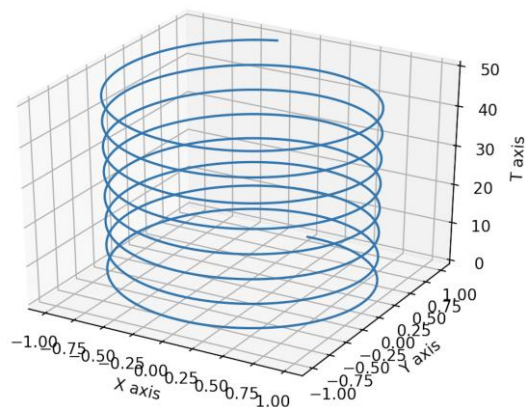


$K = 3$

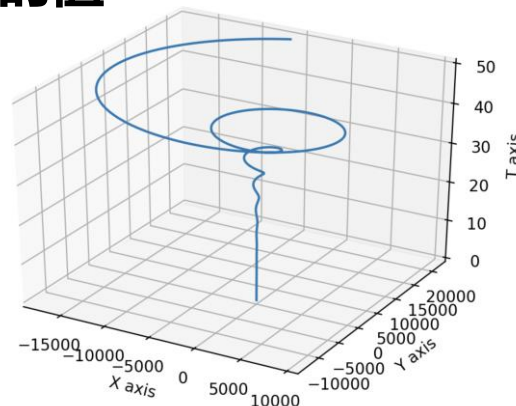
➤ 复指数信号 $f(t) = Ke^{st} = Ke^{\sigma t} \cdot (\cos \omega t + j \sin \omega t)$

K, σ 和 ω 的取值不同，复指数信号有什么不同？

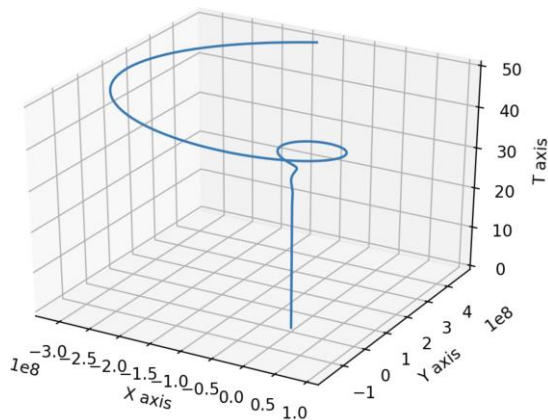
$K=1, \omega=1, \sigma$ 取不同的值



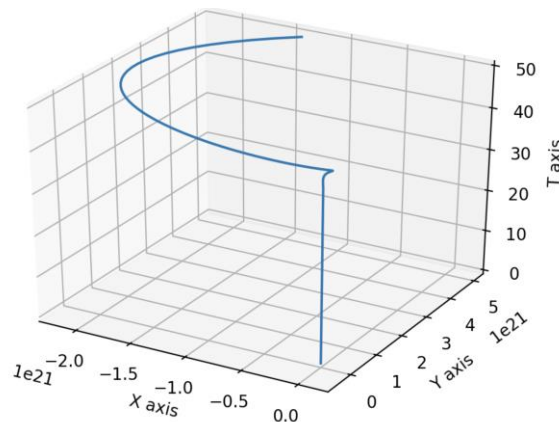
$\sigma = 0$



$\sigma = 0.2$



$\sigma = 0.4$



$\sigma = 1$

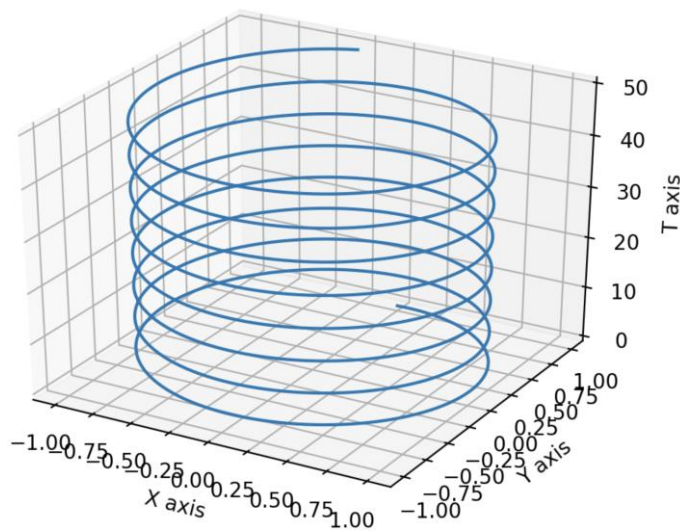
欧拉公式——应用

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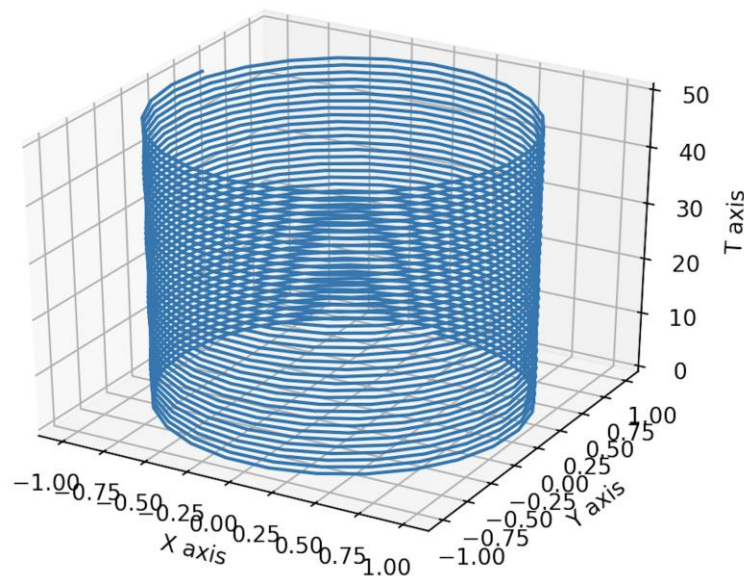
➤ 复指数信号 $f(t) = Ke^{st} = Ke^{\sigma t} \cdot (\cos \omega t + j \sin \omega t)$

K, σ 和 ω 的取值不同，复指数信号有什么不同？

$K=1, \sigma=0, \omega$ 取不同的值

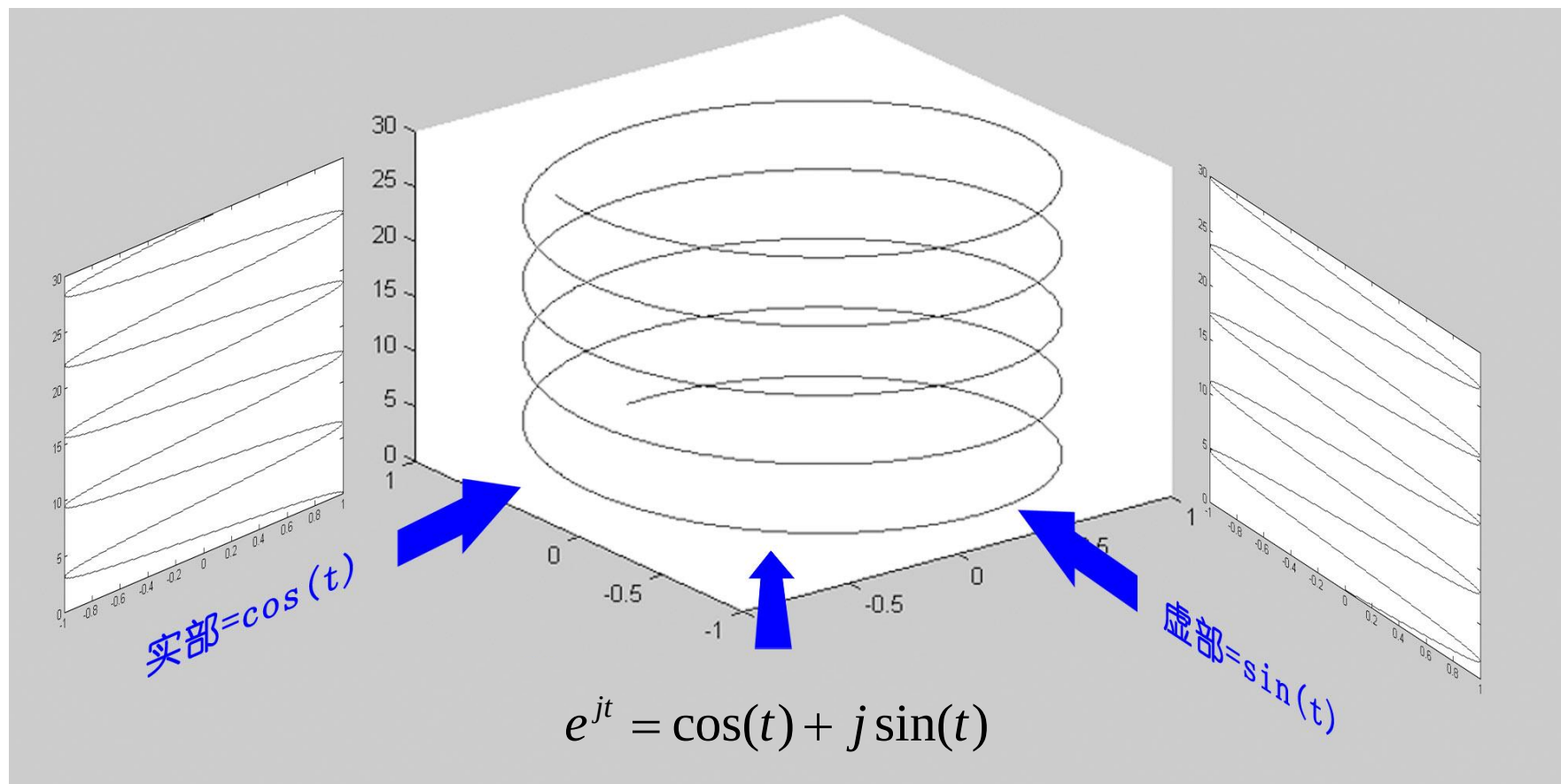


$\omega=1$

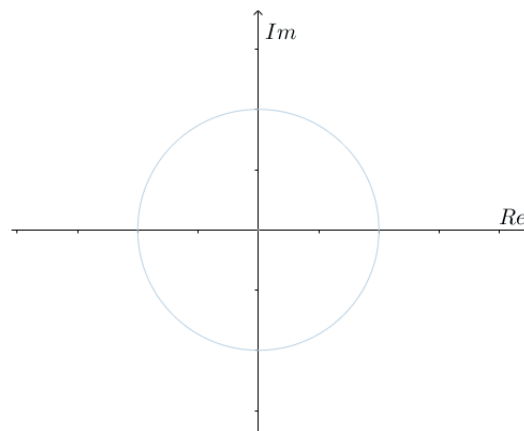
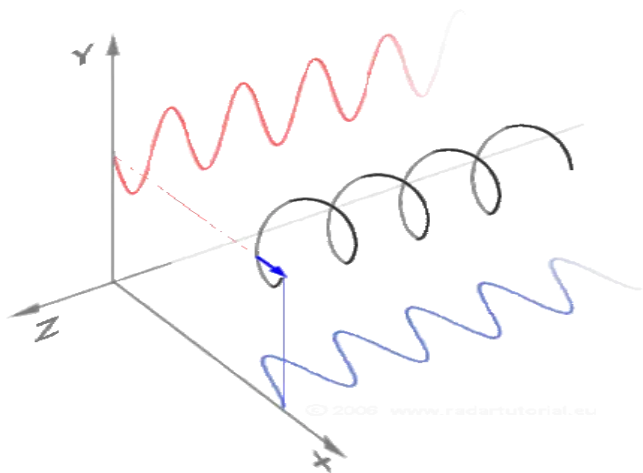


$\omega=5$

$$e^{jt}$$



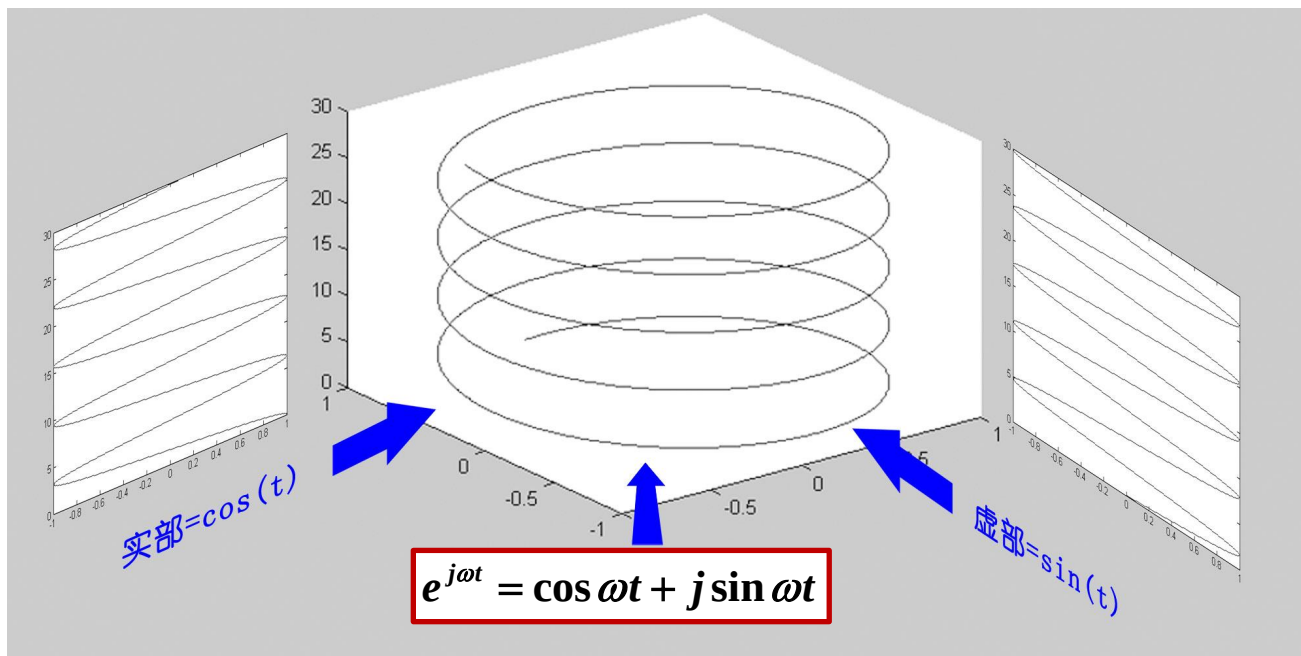
$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



$$e^{j\varphi} = \cos(\varphi) + j \sin(\varphi)$$

➤ 复指数形式傅立叶级数

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



➤ 复指数形式傅立叶级数

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

$$\cos n\omega_i t = (e^{jn\omega_i t} + e^{-jn\omega_i t}) / 2$$

$$\sin n\omega_i t = (e^{jn\omega_i t} - e^{-jn\omega_i t}) / (2j)$$

➤ 复指数形式傅立叶级数

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[\frac{a_n - jb_n}{2} e^{jn\omega_1 t} + \frac{a_n + jb_n}{2} e^{-jn\omega_1 t} \right]$$

$$f(t) = F(0) + \sum_{n=1}^{\infty} \left[F(n\omega_1) e^{jn\omega_1 t} + F(-n\omega_1) e^{-jn\omega_1 t} \right]$$

欧拉公式：

$$\cos n\omega_1 t = (e^{jn\omega_1 t} + e^{-jn\omega_1 t}) / 2$$

$$\sin n\omega_1 t = (e^{jn\omega_1 t} - e^{-jn\omega_1 t}) / (2j)$$

简写为

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

，其中：

$$F(0) = a_0$$

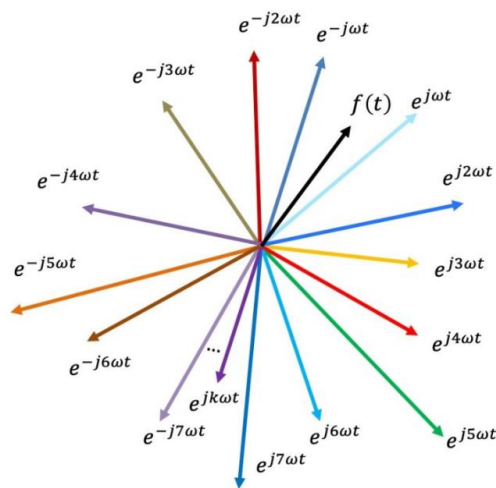
$$F_n = F(n\omega_1)$$

➤ 复指数形式傅立叶级数

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

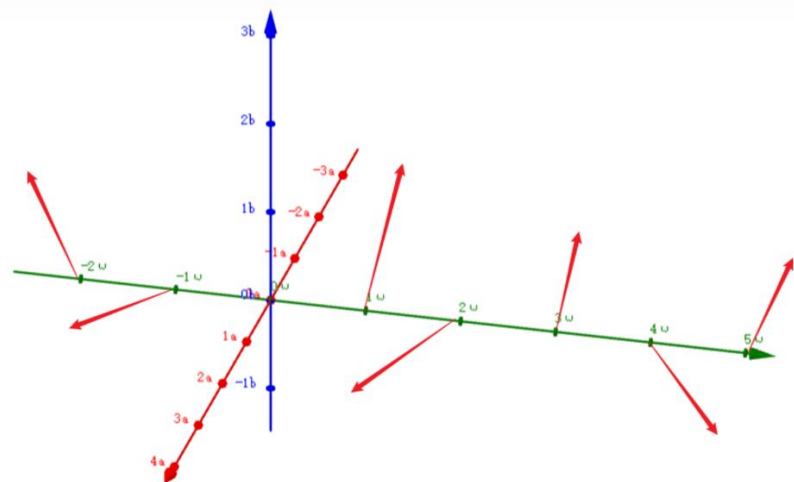
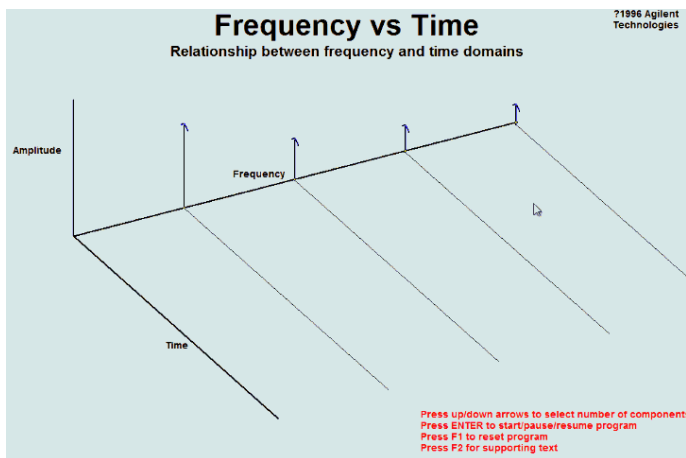
$$\begin{aligned} F_n &= \frac{a_n - jb_n}{2} = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) (\cos n\omega_1 t - j \sin n\omega_1 t) dt \\ &= \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt \end{aligned}$$

$$k_i = \int_{t_0}^{t_0+T_1} e^{jn\omega_1 t} e^{-jn\omega_1 t} dt = T_1$$



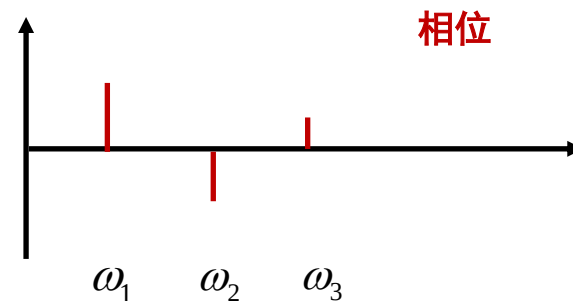
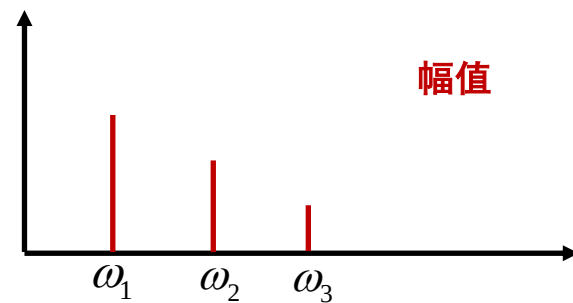
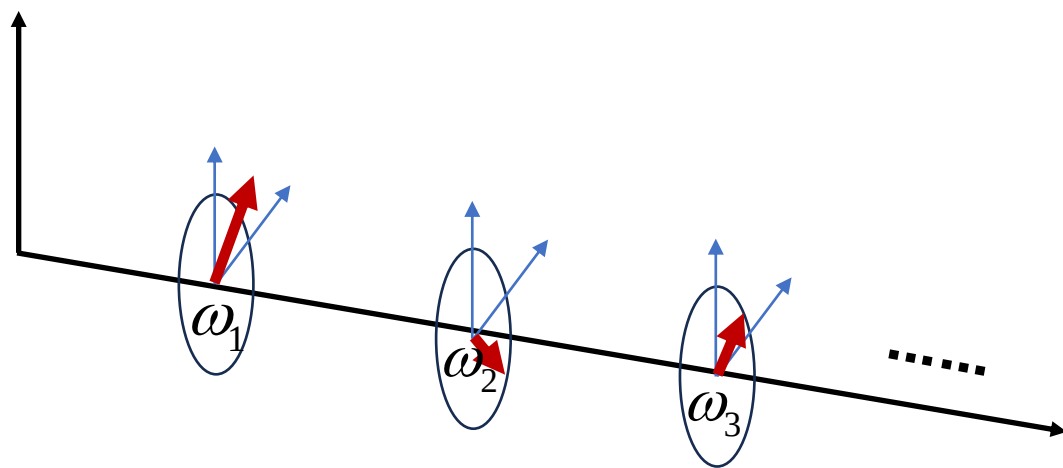
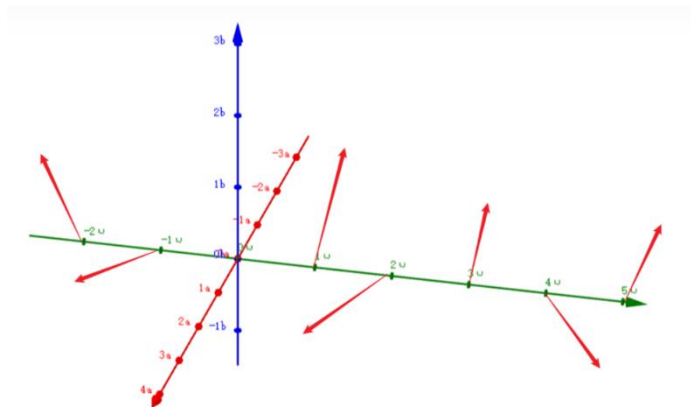
➤ 复指数形式傅立叶级数

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$



➤ 复指数形式傅立叶级数

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

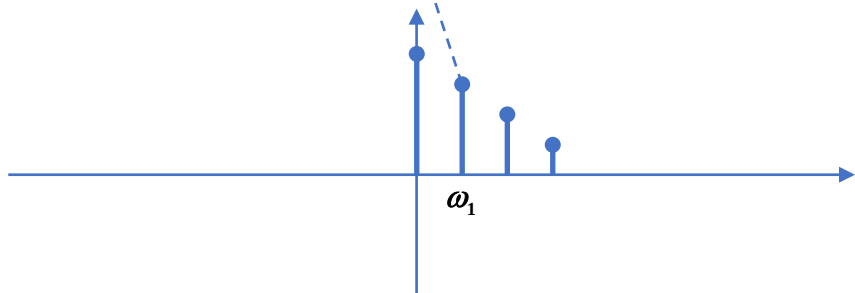


➤ 复指数形式傅立叶级数

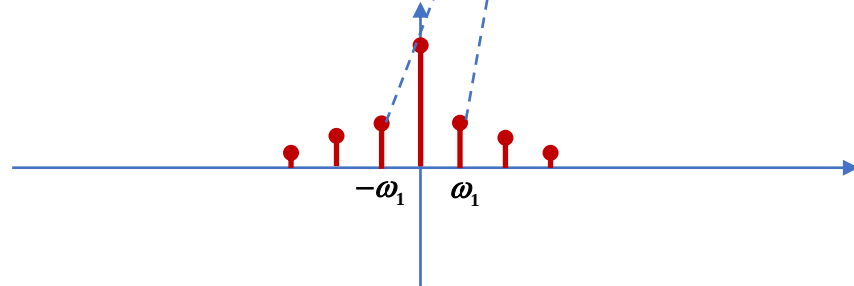
负频率的物理意义

Why?

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$
$$= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n)$$



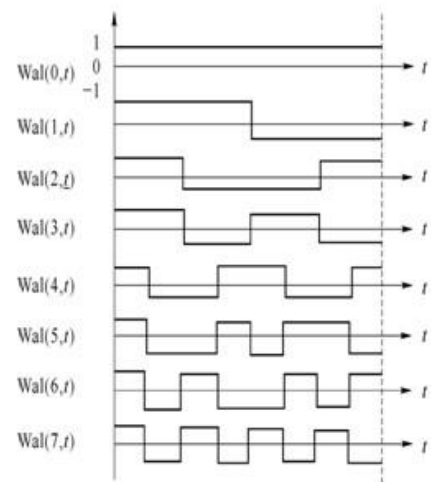
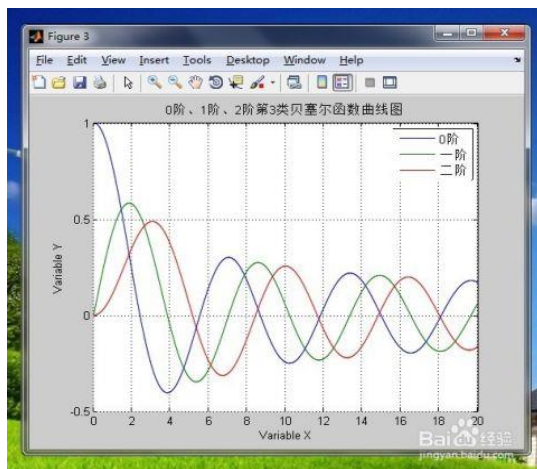
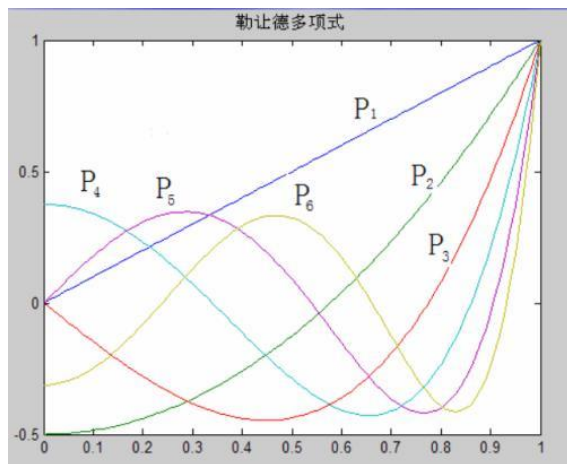
$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$



$$F_n = \frac{1}{2} c_n$$

➤ 其他形式傅立叶级数

- 常用的正交函数集的基本函数，除正弦型函数（含复指数函数）外，还有**勒让德函数**(Legendre function)、**贝塞尔函数**(Bessel function)、**沃尔什函数**(Walsh function)等。



➤ 帕斯瓦尔定理：能量守恒

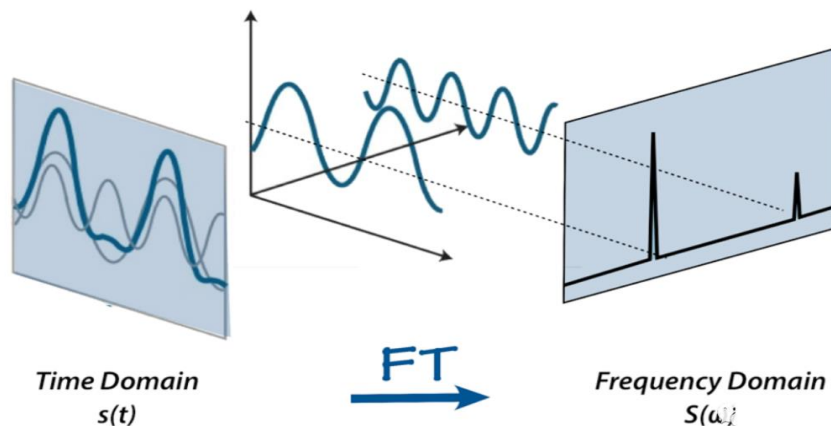
$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$

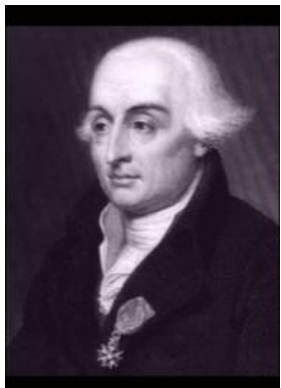
$$\int_{t_0}^{t_0+T_1} \|f(t)\|^2 dt = T_1 \sum_{n=-\infty}^{+\infty} \|F_n\|^2$$

$$\frac{1}{T_1} \int_{t_0}^{t_0+T_1} \|f(t)\|^2 dt = \sum_{n=-\infty}^{+\infty} \|F_n\|^2$$

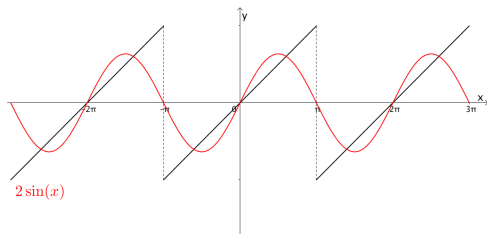


周期信号的傅立叶级数——总结

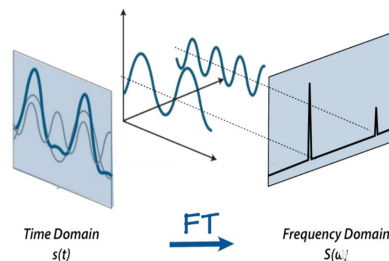
88



拉格朗日
(1736-1813)



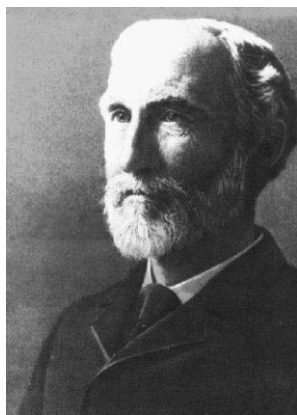
某些周期函数



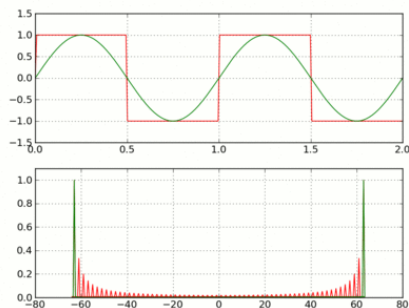
所有周期函数



傅立叶
(1768—1830)



吉布斯
(1839-1903)

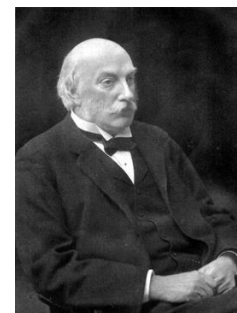


“吉布斯耳”

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=-\infty}^{+\infty} \|F_n\|^2$$

能量守恒

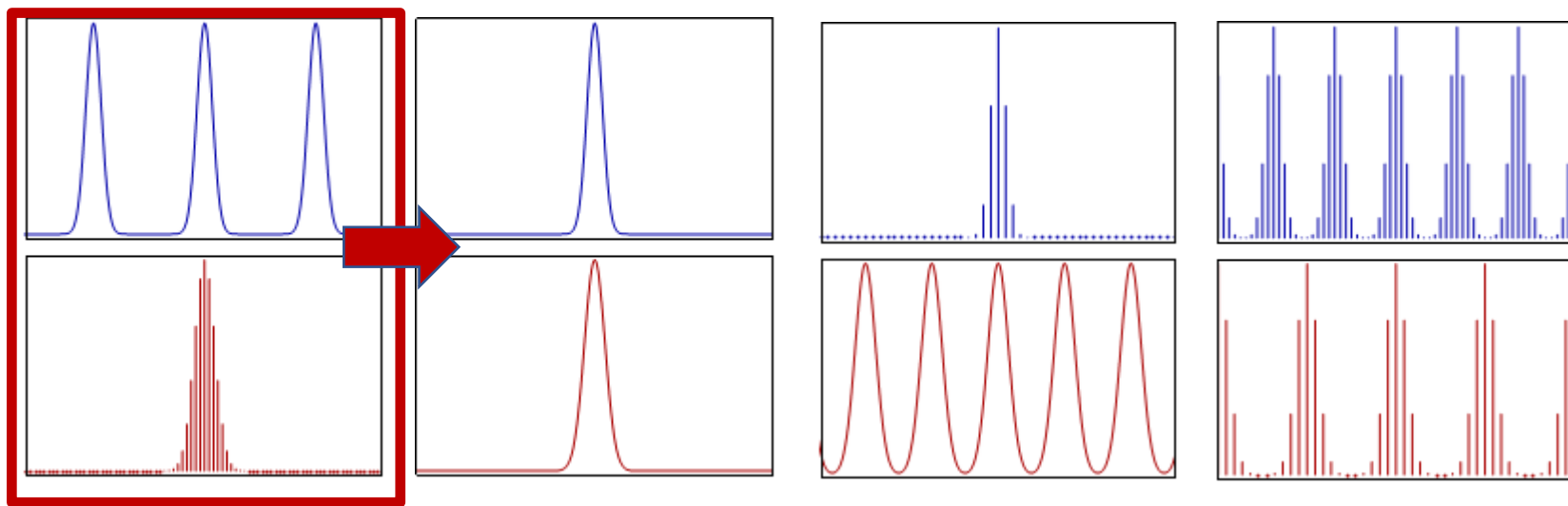


帕斯瓦尔
(1755-1836)

是结束还是开始

只能针对周期信号

非周期信号怎么办



1. 任选一种不同于课程介绍的方法证明欧拉公式

2. 证明三角函数序列

$\{1, \cos(\omega_1 t + \varphi_1), \cos(2\omega_1 t + \varphi_2), \dots, \cos(n\omega_1 t + \varphi_n)\}$

是在 $[0, 2\pi/\omega_1]$ 区间的正交函数集。

3. 求函数

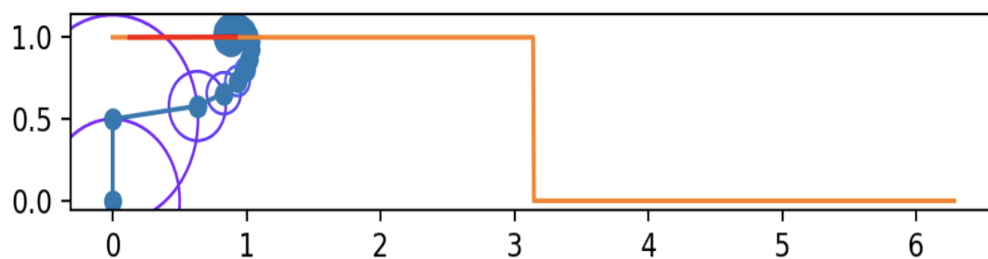
$$f(t) = \sin(t)\cos(2t) + 5\cos(3t)\sin(4t)$$

的傅立叶级数。

实验一：傅立叶级数可视化

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- 对于周期为 T_1 的周期函数 $f(t)$ ，傅立叶级数展开可以写为
$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$
- 傅立叶级数展开可以如下图构造为多个圆上运动的点 m_i 纵坐标的叠加，每个点的转动角频率固定



- 在实验一中以代码补全的形式可视化这一过程
- 可视化方波信号，选做:半圆波信号
- 环境: python3.6
- 截止日期: 10月23日